



Food and Agriculture
Organization of the
United Nations

FAO
Inclusive
Agri-food
Systems
Working
Papers
26-05

ISSN 3135-6907

Climate shocks and youth labour

Gender-disaggregated evidence from
sub-Saharan Africa

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by

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Rome, 2026

Required citation:

Kluth, J., Sitko, N.J. & Rossi, J.M. 2026. *Climate shocks and youth labour – Gender-disaggregated evidence from sub-Saharan Africa*. FAO Inclusive Agrifood Systems Working Papers, No. 26-05. Rome, FAO. <https://doi.org/10.4060/cd8526en>

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ISSN 3135-6907 (print)

ISSN 3135-6915 (online)

ISBN 978-92-5-140499-7

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Abstract

Sub-Saharan Africa (SSA) faces the dual challenge of expanding working opportunities for a rapidly growing youth population and safeguarding livelihoods from increasingly frequent extreme weather events. Despite the urgency of both issues, little evidence exists on how climate stressors affect the labour outcomes of young people in rural areas. To address this gap, we apply a double-hurdle regression model to a dataset combining socio-economic data with high-resolution climatic and geospatial information from rural areas in six SSA countries. Our results show that when rural areas are exposed to extreme heat, precipitation or droughts, young people adjust their engagement in productive labour in different ways than older adults. Moreover, these climate stressors contribute to a widening of gender gaps among youth. On the one hand, exposure to heat stress pushes young women out of productive labour, thereby widening the gap in participation between young women and young men. On the other hand, exposure to extreme precipitation and droughts is associated with a concentration of young female workers in the agricultural sector, and of young men in the non-agricultural sector. Our results highlight the importance of integrated labour and climate policies that are attentive to age and gender-specific challenges and opportunities.

Keywords: Youth labour; labour outcomes; working times; gender inequality; climate change; climate stressors; extreme weather events; rural areas; rural livelihoods; sub-Saharan Africa; Ethiopia; Malawi; Niger; Nigeria; the United Republic of Tanzania; Uganda

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Acknowledgements

This paper serves as a background document for the *Youth in Agrifood Systems Report*, launched in July 2025 by the Rural Transformation and Gender Equality Division (ESP) of the Food and Agriculture Organization of the United Nations (FAO).

The authors would like to acknowledge the contributions of Marwan Benali (ESP, FAO) and Felix Kwame Yeboah (Michigan State University) during the conceptualization phase of the paper. We are also grateful to all participants in the Youth in Agrifood Systems Consultation Workshop for their useful feedback on an earlier version of this research. During the review phase, additional valuable comments were provided by Erdgin Mane (ESP, FAO) and Marwan Benali (ESP, FAO).

Finally, we appreciate the editorial support and layout design provided by Virginie Matteredne (ESP, FAO).

Abbreviations

AMEs	average marginal effects
FAO	Food and Agriculture Organization of the United Nations
ISIMIP	Inter-Sectoral Impact Model Intercomparison Project
LSMS–ISA	Living Standards Measurement Study – Integrated Surveys on Agriculture
NAPs	national adaptation plans
NDCs	nationally determined contributions
NREGS	National Rural Employment Guarantee Scheme
OLS	ordinary least squares
SPEI	Standardised Precipitation–Evapotranspiration Index
SSA	sub-Saharan Africa
URCA	Urban–Rural Catchment Areas
VIIRS	Visible Infrared Imaging Radiometer Suite

Introduction

Climate change is driving an increase in the intensity and frequency of extreme weather events, such as extreme precipitation, heat waves, and droughts (IPCC, 2022). These events have profound impacts on agrifood systems and the livelihoods of people who depend on them (Connolly-Boutin and Smit, 2016; Thornton *et al.*, 2014). While all countries are contending with these effects, the livelihoods of people in low- and middle-income countries are particularly vulnerable. In these countries, large shares of the population derive their food and income from climate-dependent primary agricultural production and agrifood system-related off-farm labour (IPCC, 2022). People living in sub-Saharan Africa (SSA) are considered particularly vulnerable. It is estimated that 37 percent of the total population in the region are at risk of experiencing severe welfare losses because of extreme weather events (World Bank, 2024).

In addition to the development burden that climate change imposes on SSA, the region faces the additional challenge of creating livelihood opportunities for a large and growing population of young people. SSA is currently home to 224 million people aged 15 to 24 years, 65.5 percent of whom live in rural areas, where work in agrifood systems is the primary source of livelihood. By 2050, this population is predicted to grow almost 1.8-fold and reach 397 million (UNDESA, 2024). Due to a lack of economic dynamism outside of agriculture, new job creation is unable to keep pace with the number of youth entering the labour market in many SSA countries (Fox and Gandhi, 2021; Fox, Senbet and Simbanegavi, 2016). For the foreseeable future, agriculture and other work in the agrifood system will remain key sources of livelihood for youth in SSA. Thus, improving labour opportunities in the agrifood system is critical for enabling countries in SSA to achieve their objectives of reducing poverty, ending hunger, and fostering economic growth and opportunities (Christiaensen and Maertens, 2022).

In the context of SSA's large rural youth population and the growing threat posed by climate change to labour opportunities in rural areas, it is important to understand the ways in which climate stressors affect rural youth employment or work.¹ To date, there is limited

¹ The terms employment and work are often used interchangeably, but they describe different concepts. According to the International Labour Organization (ILO), employment refers any activity to produce goods or provide services for pay or profit. Work, in contrast, means any activity performed to produce goods or to provide services for use by others or for own use (ILO, 2013). In other words, work comprises employment but not vice versa. In this study, we consider the broader concept of work and use the terms work and labour to refer to it.

evidence on the effects of climate stressors on labour outcomes in SSA. The evidence that exists stems primarily from country case studies, which do not allow us to draw general conclusions on these relationships.² Moreover, youth-specific evidence is particularly scarce and, where it exists, it is primarily focused on younger age cohorts, for whom labour market entry is associated with an early withdrawal from education. For example, studies have linked floods, tropical storms and droughts to school dropouts and an increased likelihood to work among school-age children and youth in Madagascar (Marchetta, Sahn and Tiberti, 2019), Uganda (Agamile and Lawson, 2021) and several Latin American countries (Acevedo, 2016; Baez *et al.*, 2017; de Janvry *et al.*, 2006). These studies argue that households' financial hardships in the aftermath of extreme weather events drive these results. In contrast, research from Mexico and India points to a reduction in young people's likelihood to work in the face of droughts (de Janvry *et al.*, 2006; Shah and Steinberg, 2017). These authors associate their results with a contraction of labour opportunities and decreased wages resulting from post-drought economic downturns, particularly in the agricultural sector. In addition, some of these studies find that the effects of extreme weather events on labour outcomes differ between young women/girls and young men/boys (Acevedo, 2016; de Janvry *et al.*, 2006; Marchetta, Sahn and Tiberti, 2019). Though the direction of the difference varies, these results highlight that gendered norms and ascribed roles can shape the effects of extreme weather events on young people's livelihoods.

This study contributes to filling the gap of evidence on the effects of climate stressors on the labour outcomes of rural youth aged 15 to 24 years in SSA. Youth in this age group are in a transition period, characterized by the completion of education, entry into the workforce and establishment of independent livelihoods. Therefore, the ways in which they respond to and navigate the impacts of climate stressors are likely to be substantially different from younger age cohorts.

Our analysis relies on secondary individual-level socioeconomic data from six SSA countries (Ethiopia, Malawi, Niger, Nigeria, the United Republic of Tanzania and Uganda). To identify the survey population's exposure to climate stressors, we match this data in both space and time with georeferenced information on incidents of extreme heat, extreme precipitation and droughts. Our empirical strategy is based on a double-hurdle regression model, which is implemented in two steps, thereby considering both the extensive and the intensive margin of work. In the first step, our outcomes are dummy variables denoting whether individuals worked in the week preceding the interview, either in general or in the agricultural or the

² To our knowledge, the study by Nico and Azzarri (2024) is the only multi-country analysis on the topic, but it does not examine age-related differences and is not specific to rural areas.

non-agricultural sectors. In the second step, conditional on having worked in general or in one of the two sectors, we examine the labour hours dedicated to the respective activities. To elicit differences in outcomes between younger and older people, as well as between young women and young men, our models include interaction terms between exposure to climate stressors and individuals' age or gender.

Our results demonstrate that younger people react to climate stressors differently than older people, but the direction and statistical significance of these effects depend on the stressor and sector considered. We find that rural societies rely more on the labour of young men when exposed heat stress. Young women, in contrast are less likely to engage in productive labour when exposed to extreme heat, thus leading to an increased gender gap in the likelihood to work among youth. Under exposure to extreme precipitation, older adults contribute relatively more labour, particularly in the agricultural sector. At the same time, when extreme precipitation or droughts occur, young women work relatively more than young men. Under both stressors, however, young women are more likely than young men to engage in agricultural activities, and less likely to work in the non-agricultural sector. Therefore, extreme precipitation and droughts open up gender gaps in participation by sector, which do not exist in the absence of climatic shocks.

The remainder of the article is structured as follows. Section 2 outlines our conceptual framework for how climate stressors affect labour outcomes, depending on individuals' age and gender. Section 3 describes the data sources, construction of variables and empirical strategy. Section 4 presents descriptive and empirical results. Section 5 discusses these results and their policy relevance, outlines the limitations of the study, and concludes.

Conceptual framework

Our conceptual framework draws on existing evidence to hypothesize how exposure to climate stressors may influence labour outcomes in rural areas, and how young people's experiences may differ from those of adults. Moreover, we recognize that youth are not a homogenous group, but rather that other social identities are likely to strongly influence how they respond to climate stressors. Of particular importance are the ways in which gendered norms may contribute to differences between the experiences of young women and young men.

In rural SSA, labour opportunities are often tied to agricultural production. It is estimated that in 2019, 53 percent of the employed population in Africa,³ roughly 247 million people, were working in agrifood systems, with the vast majority engaged in agricultural employment (Davis *et al.*, 2023). Engagement in agrifood systems is particularly important for women in SSA, where estimates show that 66 percent of working women are engaged in agrifood system work, compared to 60 percent of men. This discrepancy is likely driven by differences in educational attainment between women and men, and the prevalence of gendered norms that restrict women's access to formal and full-time employment outside of the household (FAO, 2023).

Climate change is having an oversized impact on agricultural production in SSA, with serious repercussions for agricultural producers and people who rely on agrifood systems for their livelihoods. Estimates show that since 1961, agricultural productivity has declined by 34 percent due to climate change in Africa, more than in any other region (IPCC, 2022). In addition to slow-onset changes in climatic conditions, the increased frequency and intensity of extreme weather events, such as heat stress, extreme precipitation, and droughts, have devastating effects on crop yields and livestock production in the region (Lobell *et al.*, 2011; Lobell and Gourdji, 2012; Thornton *et al.*, 2009).

Reductions in agricultural production pose an immediate threat to food security in SSA (Wheeler and von Braun, 2013), while also affecting labour opportunities in the agrifood system and beyond. Decreased agricultural production reduces demand for agricultural labour and associated work in the agrifood system, such as in trading and processing of food (Gray, Taraz and Halliday, 2023). As work opportunities in agrifood systems decline, people

³ Estimates are for the African continent and are not available for sub-Saharan Africa.

depending on them reduce their expenditures, thereby causing financial constraints and reduced employment in businesses outside of the agrifood system as well (Acevedo *et al.*, 2020; Escalante and Maisonnave, 2022; Jessoe, Manning and Taylor, 2018; Santangelo, 2019). At least in the short run, wage reductions resulting from decreases in labour demand amplify this effect (Banerjee, 2007; Mueller and Quisumbing, 2011; Oliveira, Palialol and Pereda, 2021). In addition, workers whose assets or livelihoods have been destroyed may be willing to work for a lower pay, thereby adding further downward pressure on wages (Banerjee, 2007). At the same time, wage differentials can guide people's decisions on how much time to spend working in which sector. For example, evidence from Bangladesh shows that while both agricultural and non-agricultural wages decreased during floods, the effect was stronger in the agricultural sector (Mueller and Quisumbing, 2011).

Age is likely to be a strong mediating factor in the relationship between exposure to climate stressors and labour outcomes in rural areas of low- and middle-income countries. Young people in these contexts face various constraints when it comes to protecting their livelihoods from climatic stress. Young workers are concentrated in vulnerable forms of labour, such as own-account work, contributing family work and paid temporary work (ILO, 2024a). Even when youth work outside their households, their labour arrangements are mostly informal and can, thus, quickly be dissolved when labour demand decreases. In addition, young people have less job experience than their older coworkers and are, therefore, often the first to lose their jobs in times of crisis (ILO, 2024a).

Self-employed youth have a higher degree of independence. However, as with other jobs, their young age means that they possess limited knowledge about how to adapt if shocks occur. In the context of primary agricultural production, farms with better climate-adaptive management practices are better able to maintain production levels and labour demands in the face of climate stressors than those with limited climate-adaptive management. Evidence shows that farmers' likelihood of adopting climate-adaptive practices increases with years of experience (Abid, Schneider and Scheffran, 2016; Aryal *et al.*, 2020; Elahi *et al.*, 2022; Gebrehiwot and van der Veen, 2013) as well as with the number of climate shocks experienced in the past (Dassanayake *et al.*, 2018; Makate *et al.*, 2022). Although there are no corresponding studies, these results are likely relevant to the non-agricultural sector too. In addition, the capacity of self-employed youth to protect their livelihoods from shocks is compromised by limited access to land and labour, as well as to physical, financial and social capital (Chakravarty, Das and Vaillant, 2017; FAO, 2024).

Yet, some factors may bolster resilience to climatic shocks among youth. From a biological perspective, people under the age of 40 are better able than older adults to handle heat stress when engaging in high-intensity physical work. Importantly, men's heat resistance is generally higher than women's (ILO, 2024b). In addition, young people have higher average

levels of education than older generations and are better able to leverage digital technologies to access information (Edmunds and Turner, 2005). Both factors may enable youth to access a larger range of labour options, particularly outside of the agricultural sector. Lastly, economic hardships can drive young people to be resourceful and innovative, thereby increasing their openness to explore different jobs, learn new skills or try advanced technologies. For example, studies show that young farmers are more likely to adopt drought-tolerant seed varieties (Kogo *et al.*, 2022) and to migrate and seek non-farm jobs (Keshavarz and Moqadas, 2021).

For these reasons, we expect that the impacts of extreme weather events on labour outcomes are likely to differ between younger and older people. However, these differences may vary based on the type of extreme event and sector of work (e.g. agriculture vs non-agricultural).

The constraints and opportunities described above likely also differ between young women and men. Young women face discriminatory norms and customs that make it more difficult for them to pursue remunerative livelihoods. For example, gender norms place the brunt of the burden for household and care work on young women, thus leaving limited time for productive work. In many contexts, gender norms also restrict young women's mobility outside their households, and deem certain jobs unsuitable for them (Chakravarty, Das and Vaillant, 2017). As a result, young women are overrepresented in vulnerable labour categories and have to reconcile productive work with unpaid household duties, thereby further reducing their capacity to cope with shocks (Chakravarty, Das and Vaillant, 2017; FAO, 2023; ILO, 2024a). Young women are also less able to capitalize on youth-specific advantages in the context of stresses and shocks. For example, young women tend to have lower educational attainment and limited access to information compared to their male peers (Chakravarty, Das and Vaillant, 2017).

Studies from Madagascar and India show that young women's labour outcomes are more negatively affected by droughts than those of young men, both in the agricultural and in the non-agricultural sector (Afridi, Mahajan and Sangwan, 2022; Dessy *et al.*, 2020). These findings are driven by gender norms that prevent young women from capitalizing on income-generating activities that would require them to work away from the household. Crucially, while young men can opt to migrate in the aftermath of a drought, this coping strategy is often not available to young women.

Several studies that examine gender differences in the relationship between climate stressors and labour outcomes among adults have produced similar results. A recurring finding is that exposure to extreme weather events widens gender gaps to the disadvantage of women in terms of the overall propensity to engage in productive labour (Acevedo, 2016;

Afridi, Mahajan and Sangwan, 2022; Arceo-Gómez, Hernández-Cortés and López-Feldman, 2020; Moshoeshe, Nchake and Tiberti, 2021; Ngepah and Conselho Mwiinga, 2022), as well as in terms of working times (Afridi, Mahajan and Sangwan, 2022; Akter, 2021; Shayegh and Dasgupta, 2024). Moreover, when faced with climate stressors, women tend to be less able than men to diversify their livelihoods towards non-agricultural sources (Afridi, Mahajan and Sangwan, 2022; Feng, Liang and Yan, 2025), or to engage in self-employment (Acevedo, 2016) or wage labour (Maitra and Tagat, 2024).

However, while the literature frequently finds climate stressors to exacerbate gendered labour disparities, this is not always the case (Baez *et al.*, 2017; Nico and Azzarri, 2024). Similarly, evidence on the effects of weather events on the gender wage gap shows different results depending on the type of event, the sector, and the role of the female and the male labour in a given production technology (Afridi, Mahajan and Sangwan, 2022; Mahajan, 2017; Shayegh and Dasgupta, 2024).

Finally, individual-level labour allocation can be the outcome of collective decision-making among household members (Vermeulen, 2002). In other words, the effect of climate stressors on youth's and women's labour outcomes could be influenced by their role in the household and their household's composition. For example, evidence shows that heat affects female agricultural labour supply differently depending on the gender distribution among household members (Lee *et al.*, 2021). Thus, *a priori* we expect that extreme weather events will affect young women and men differently, but the direction of these differences may not be uniformly regressive for women.

Materials and methods

Socioeconomic data

Data sources

We obtain socio-economic information at the household and individual level from six sub-Saharan African countries included in the World Bank's Living Standards Measurement Study – Integrated Surveys on Agriculture (LSMS–ISA, see Table 1 for the countries, surveys, waves and sample sizes). LSMS–ISA surveys comprise comprehensive modules on household members' participation in the labour force. In the selected countries and waves, the questions are similar in terms of phrasing and recall periods. We rely on survey waves from 2019 or earlier to avoid the use of data that has been influenced by the COVID-19 pandemic. We only use the data from working-age individuals, defined as being 15 to 64 years old, residing in rural areas. Depending on the model and outcome variable, the sample size ranges between about 2 000 and 56 000 observations.

Table 1. Countries, surveys and waves for analysis

Country	Survey	Wave	Sample size
Ethiopia	Ethiopia Socioeconomic Survey	2019	6 478
Malawi	Integrated Household Survey	2017	21 924
Niger	National Survey on Household Living Conditions and Agriculture	2014	2 601
Nigeria	General Household Survey	2019	9 662
United Republic of Tanzania	National Panel Survey	2013	8 899
Uganda	Uganda National Panel Survey	2016	6 145

Source: authors' own elaboration

Construction of outcome variables

Our outcome variables measure individual-level working times per week, typically in the 7 days prior to the interview.⁴ The first variable captures all hours worked in respondents' main and secondary jobs. The second variable considers only hours worked in the agricultural sector, i.e. in primary production on the household's own farm or as wage/casual labour on someone else's farm. The third variable considers hours worked in the non-agricultural sector.⁵ In line with common practice in labour economics, we cap the total weekly working time at 84 hours per week and replace larger values with this maximum value (Quiñones *et al.*, 2009).

Climate data

Data sources

The analysis of heat stress and extreme precipitation is based on the WFDE5 dataset, which we accessed from the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP) (Cucchi *et al.*, 2020). From this dataset, we extracted information on the daily maximum temperature and the daily total rainfall from 1979 to 2019 at a 0.5°×0.5° spatial resolution. Data for the analysis of droughts is taken from the Global SPEI database, which provides monthly precipitation and evapotranspiration data since 1901 at a 0.5°×0.5° spatial resolution (Vicente-Serrano *et al.*, 2022).

In addition, we obtain several other climatic and geospatial variables to account for factors that may influence individuals' labour decisions in the face of climatic stress. Specifically, we use the ERA5-Land dataset (Muñoz-Sabater *et al.*, 2021) to compute 30-year means and coefficients of variation for temperature and precipitation. Moreover, our estimations control for population density from the WorldPop dataset (WorldPop, 2018), night-time lights from the Visible Infrared Imaging Radiometer Suite (VIIRS) (Elvidge *et al.*, 2021), and a measure of travel time to cities of different sizes from the Urban–Rural Catchment Areas (URCA) dataset (Cattaneo, Nelson and McMenomy, 2021).

⁴ In the Niger, the survey asks for the number of hours that respondents *usually* worked in the main/secondary job they had in the preceding week. See Appendix A1 for a detailed description of how the variables were constructed in each country.

⁵ We would have liked to define the sectorial variables according to the agrifood systems (AFS) classification, i.e. differentiating between primary agricultural production AFS-related processing and services, and non-AFS-related processing and services. However, this classification requires ISIC codes of at least 2 digits (Davis *et al.*, 2023), which are not available in the surveys from Ethiopia and Nigeria.

Construction of explanatory climate variables

We proxy heat stress with the number of days per year with a maximum temperature above the 99th percentile of the historical (1979–2009) distribution of daily maximum temperature. Therefore, given the interview date of the household, h , in the administrative division, l , the heat stress indicator is:

$$H_{hl} = \sum_{d=1}^{365} H(T_{dl} - T99_l) \quad (1)$$

where T_{dl} is the daily maximum temperature and $T99_l$ is the historical 99th percentile. $H()$ is the Heaviside step function, which takes the value of 1 if the daily precipitation is above the threshold, and 0 otherwise.

Similarly, we define extreme precipitation as the number of days with precipitation above the 95th percentile of the historical (1979–2009) distribution of daily precipitation. Therefore, the indicator is:

$$F_{hl} = \sum_{d=1}^{365} H(P_{dl} - P95_l) \quad (2)$$

where P_{dl} is precipitation on day d and $P95_l$ is the historical 95th percentile.

Droughts, in contrast, are identified using the Standardised Precipitation-Evapotranspiration Index (SPEI), which measures the difference between precipitation and evapotranspiration in a given location and period of time and compares it with the long-term historic record (since 1901). The SPEI is standardized using a log-logistic distribution.

Therefore, the difference between the SPEI of the reference period and the long-term record is measured in standard deviations. In this analysis, we use the SPEI6 indicator, which is calculated over a six-month period. To identify extreme droughts, we construct a binary variable that takes a value of 1 if in any of the six months preceding the interview the SPEI6 < -1.5 standard deviations:⁶

$$D_{hl} = \begin{cases} 1, & \text{if } SPEI_{dl} < -1.5 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

⁶ This approach allows us to capture one entire year before the interview, because the SPEI6 of the sixth month goes back to the 12th month before the interview.

This procedure captures very dry and extremely dry conditions.

We use local yearly extreme values to define the thresholds of heat stress, extreme precipitation and droughts. This approach ensures that spatial comparisons are meaningful, as they measure anomalies with the same probability of occurrence across space (Tank, Zwiers and Zhang, 2009; Zhang *et al.*, 2011).

Matching the socioeconomic and climate data

While the climatic and spatial control data is provided in grid form with a high degree of spatial resolution, the location of the survey households is not available at the same level of granularity (see Table 2 for the lowest available administrative level by country). To spatially associate the different data sources, we assign all the available grid cell values to the administrative division with which they overlap, weighted by the proportion of the cell falling within the respective division (Flatø, Muttarak and Pelsler, 2017; Letta, Montalbano and Pierre, 2022). This approach results in a single value for all households within the same administrative division. We apply the same strategy to match the survey data with the VIIRS and WorldPop datasets. We aggregate the night-time light data using the mean and the population data using the sum of an administrative division. To obtain population density, we divide the administrative division's total population by its surface area.

Table 2. Lowest administrative level available in the household surveys

(1) Country	(2) Adm. div. level	(3) Adm. div. name	(4) No. of adm. div. in survey (excl. urban only)	(5) No. of adm. div. in analysis ⁷	(6) Median surface area (km ²) of adm. div.	(7) Share of adm. div. smaller than 3 000 km ²	(8) Median population of adm. div.
Ethiopia	3	Woreda (district)	228	194	1 099	84%	142 848
Malawi	2	Traditional Authority (sub-district)	209	209	342	99%	54 832
Niger	2	Department	33	14	7 460	14%	543 398
Nigeria	2	Local Authority (district)	286	275	819	90%	240 513
United Republic of Tanzania	3	Wodi (ward)	761	724	121	98%	12 101
Uganda	1	District	104	53	3 300	47%	619 977

Source: authors' own elaboration

To temporally associate the exposure to weather events with the survey data, we use the interview dates as a reference point and assign each household with the climatic data of the 12 months preceding the interview. We choose 12 months as it is the reference period in the surveys. Since URCA is only available for 2015, we use this year for all the surveys. For night-time lights and population, we use data from the year of the survey implementation.

Table 2 also shows a substantial degree of variation in terms of spatial granularity across countries. The median size of administrative units varies from 121 km² in the United Republic of Tanzania, with a median population of 12 101 persons, to 7 460 km² and 543 398 persons in Niger (see columns (6) and (8)). This characteristic of the data effectively reduces

⁷ Two factors lead to “losses” of administrative divisions. First, in Niger, Nigeria and Uganda, there are substantial non-response rates to key sections of the questionnaire, such as the labour module. Second, for all surveys, some administrative divisions cannot be matched with the climate data. This problem is particularly relevant in Niger, and to some degree also in Ethiopia, United Republic of Tanzania and Uganda.

variation across countries. However, in four out of the six countries, over 80 percent of the administrative units are smaller than the size of the grid cells in the climate data, which in the region amounts to $\pm 3\,000\text{ km}^2$ (column (7)). Moreover, all surveys except Niger include a large number of administrative divisions (column (5)). Therefore, and given the large number of individuals in the sample and the different interview dates which determine the reference period for the climate variables, our data allows for sufficient spatial and cross-country variation to support our identification strategy.

Empirical strategy

This paper discusses the effects of climate stressors on the labour outcomes of rural individuals, and measures how these effects differ between younger and older working-age adults, as well as between young women and men. We define working-age individuals as being 15 to 64 years old, and youth as being 15 to 24 years old. Our first set of models considers all working-age individuals, thereby allowing us to examine the interaction between climate stressors and young age. Subsequently, the second set of models restricts the sample to young individuals only to analyse how the effects of climate stressors on labour outcomes differ between young women and men.

Weekly labour time does not follow a normal distribution. Instead, we observe a mass point at zero for all individuals who do not work, and positive values for all individuals who do. To account for this mixed discrete-continuous nature of our outcome variables, we apply a double-hurdle or two-part model (Belotti *et al.*, 2015; Cragg, 1971). The first part is a probit regression model, which estimates an individuals' likelihood to have worked a positive number of hours in the 7 days preceding the interview. The second part restricts the sample to observations with positive working times and fits an ordinary least squares (OLS) regression model. In this second step, the outcome variable is log-transformed to deal with the right-skewedness of its distribution.

Previous studies have applied double-hurdle models to analyse labour supply and demand (e.g. Benali, Brümmer and Afari-Sefa, 2018; Matshe and Young, 2004; Rao and Qaim, 2013). Double-hurdle models present an alternative to the otherwise frequently used Tobit and Heckmann models. However, as opposed to Tobit models, double-hurdle models allow the decision whether to participate in the labour market and the decision on the quantity of labour supplied to be generated by two different processes. Moreover, in contrast to Heckman models, double-hurdle models do not treat zeros as censored values, but as the result of an individual's choice not to participate in the labour market. Reasons for such decisions can be personal preferences, inadequate qualifications or disabilities. Moreover,

individuals may choose not to work given the prevailing economic climate and earnings potential (Matshe and Young, 2004; Rao and Qaim, 2013).

We run three separate double-hurdle regressions, one per climate stressor (C), for all six combinations of samples (all working age and youth only) and outcome variables (weekly working hours in total, in agriculture and in non-agriculture). To elicit the differential impacts of these stressors, we interact the respective stressor variable with a subgroup indicator (S), which denotes either age or gender. Previous papers have applied this approach to study heterogeneous effects of climate shocks and other factors on labour outcomes (e.g. Egger, Arslan and Zucchini, 2022; Nico and Azzarri, 2024).

The following equation formalizes the estimation model:

$$Y_{ihl} = \alpha + \beta_1 S_{ihl} + \beta_2 C_{hl} + \beta_3 C_{hl} \times S_{ihl} + \gamma_1 I_{ihl} + \gamma_2 H_{hl} + \gamma_3 G_l + \delta_c + \varepsilon_{ihl} \quad (4)$$

The outcome variables, denoted by Y_{ihl} , are measured at the individual level i , which is nested within the household h and administrative division l . Depending on the specification, S_{ihl} indicates either a person's age or gender. Both variables take the value of 1 if the individual is young or female, and 0 otherwise. The coefficient β_1 measures the effect of belonging to a given subgroup on the outcome variable, regardless of exposure to climate stress.

Depending on the specification, C_{hl} denotes one of the three climate stressors, i.e. heat stress, extreme precipitation or drought. These variables are measured at the level of the administrative division l , but vary according to the interview date of the household h . The coefficient β_2 captures the effect of the respective climate stressor on the outcome variables of the reference group, i.e. older working-age individuals or young men.

Our key coefficient of interest is β_3 , which measures how the effect of a given climate stressor differs according to the subgroup. A statistically significant coefficient indicates that the stressor affects the respective subgroup differently than the reference category. Take the example of total labour time and assume that both β_2 and β_3 are negative and statistically significant. This result means that the climate stressor leads to a general decrease in working time, but that this decrease is stronger for young/female individuals compared to their older/male counterparts. Accordingly, a positive coefficient β_3 signals that the subgroup is less affected than the reference group or even increases weekly labour time.

Vectors I_{ihl} and H_{hl} represent covariates at the individual and household level, respectively. At the individual level, we control for a person's age (level and squared term), completed

years of education, an indicator for whether they currently attend school, and an indicator for whether they are married. In the model exploring differences between youth and adults, I_{ihl} also includes a dummy for gender. The household-level covariates are household size, the gender of the household head, the average age of all household members (level and squared term), the average years of education of all household members, the share of dependent members, the ratio of female over male household members, the share of household members who have a job, an indicator denoting whether the household receives any form of social protection, and an indicator denoting whether the household is involved in agricultural activities. Furthermore, we include an indicator which takes the value of 1 if the household is poor. This variable is derived from a household-level multidimensional poverty index following the methodology developed by Alkire and Foster (2011) as closely as data availability allows. The index accounts for seven dimensions: communication, assets, dwelling characteristics, access to water, electricity, sanitation, and education. Households are considered “poor” if they fall within the lowest quartile of a country’s poverty index distribution. To some extent, these household covariates allow us to control for the role that intra-household dynamics play in individual-level decision-making.

Lastly, vector G_l accounts for the following geospatial characteristics: population density as a proxy for remoteness, night-time lights as a proxy for economic development, and URCA as an indicator for proximity to markets. G_l also accounts for location-specific general climatic conditions using the long-run means and coefficients of variation of temperature and precipitation.

All models include country-fixed effects (δ_c). We assume that the errors, ε_{ihl} , follow a normal distribution, and allow for correlation of errors within households by clustering the variance-covariance at the household level.

If the outcome variable is total weekly working hours, we run the models for all individuals of working age and young age, respectively. In contrast, the models for hours worked in agriculture and non-agriculture consider only those who worked for at least one hour during the recall period. Thus, the sectorial models analyse participation in agriculture and non-agriculture, given that the respondent has decided to work.

Since the coefficients produced for the first and second step are not straightforward to interpret, we calculate average marginal effects for each of the two steps (conditional average marginal effects, AMEs). Moreover, we apply Duan’s smearing retransformation (Duan, 1981) as well as bootstrapping of standard errors (Belotti *et al.*, 2015) to account for the non-normal distribution of residuals in the first step and the log-transformation of the dependent variables in the second step.

Sources of bias

Given that our empirical approach is of a non-experimental nature, we cannot make rigorous causal inference about the impact of climatic stressors on rural individuals' labour outcomes. Nevertheless, several features of our estimation model make us confident that the identified relationships are robust. First, the indicators for climate stressors are constructed from an external and independent data source, rather than based on respondents' self-reported experiences. Therefore, our dependent variables are strictly exogenous to the independent ones. Further, we can rule out any risk of reverse causality, as individuals are unable to influence the weather and climate to a significant degree.

Second, we control for a wide range of household-level and geospatial characteristics to reduce the risk of omitted variable bias. For example, geographic location is associated with exposure to extreme weather events, as certain orographic and environmental characteristics make the occurrence of heat and precipitation extremes more common in some areas than in others. At the same time, geographic location correlates with the availability of livelihood opportunities, as well as with rural peoples' capacities to generate income. To avoid confounding the effects of historical climatic trends with those of recent extreme events, we use location-specific thresholds to define our climatic dependent variables and control for the long-run means and coefficients of variation of temperature and precipitation. Other structural differences across locations are captured by the remaining location-level variables.

A third concern for bias is the self-selection of households into climatically more stable locations. For example, households that are wealthier or have better access to non-agricultural income sources might be able to migrate if they face severe climatic stress in their home regions. Therefore, a positive relationship between exposure to climate stressors and labour outcomes exists even if climate stressors do not directly impact labour decisions. We factor out the effect of such wealth differences between households by including a wealth indicator, as well as the household size and the average age and education levels of household members in all equations. However, the information provided in the household surveys does not allow us to control for migration behaviour.

In addition to migration, there are other potentially important unobservable or unmeasured factors that could distort our results. For example, while we can account for population density and remoteness, we are not able to capture household-level access to and use of information, e.g. weather forecasts from mass media or extension services. Moreover, we cannot factor in the potentially confounding effects of changes in wage levels.

Lastly, we acknowledge that our outcome variables and the variables for exposure to climate stressors are measured over different time frames (one week vs one year before the interview). Thus, we assume that exposure to an extreme weather event at any moment in the year preceding the interview influences the labour outcomes in the reference week, and that this influence does not vary depending on the exact timing of the event.

Results

Descriptives

Table 3 shows descriptive statistics for the full sample of working-age individuals and by age group. Overall, 74 percent of the working-age population have a job, and 65 percent worked for at least one hour in the week preceding the interview. The average weekly labour time among workers amounts to about 26 hours, considering both the primary and secondary job. The likelihood to work and weekly working times are both significantly higher for adults than for youth. Among youth, 62 percent have a job and 52 percent worked in the week before the interview. On average, these youth worked for 22 hours per week. Among adults, 81 percent have a job, 73 percent worked in the last week and workers' average weekly working time amounts to 28 hours.

Across both age groups, participation in agriculture is significantly higher than participation in non-agriculture. Specifically, 89 percent of working youth engage in agriculture but only 16 percent work in non-agriculture. For working adults, participation in agriculture and non-agriculture stand at 83 percent and 29 percent, respectively. However, for both age groups, working times in non-agricultural activities are significantly longer than in agricultural ones. In both sectors, more individuals work at the household level than outside the household.

Overall, young workers are more likely than older workers to engage in agriculture, both at the household level and in wage or casual labour. In contrast, adult workers are more likely than youth to engage in the non-agricultural sector, both within and outside the household. For all activities except non-agricultural wage or casual labour, working times are significantly shorter among youth than among adults.

In terms of exposure to climate stressors, we observe statistically significant differences between youth and adults, but the magnitude of these differences is very small. On average, the sampled individuals experienced 5 days of heat stress and 6 days of extreme precipitation per year. Moreover, 20 percent of the sample has been exposed to drought.

In line with the literature reviewed earlier, young people in our sample have completed more years of education than adults (6.6 vs 5.9 years on average). Moreover, youth are 5 percentage points less likely than adults to reside in a poor household. However, youth's households are more likely to be female-headed and tend to be larger and younger on average, though with fewer dependent members and a higher share of working members.

Table 4 shows descriptive statistics for the sub-sample of youth, disaggregated by women and men. The figures reveal that young men are significantly more likely than young women to have a job (65 percent vs 58 percent) and to have worked in the week preceding the interview (56 percent vs 48 percent). Moreover, conditional on having worked in the week before the interview, young men work on average 5 hours more per week than young women.

Working young women and men are roughly equally likely to engage in agriculture (89 vs 90 percent) and non-agriculture (16 percent for both). However, in both sectors, young women are more likely than young men to work at the household level, and less likely to work outside the household. Young women's working times are significantly lower than young men's in all activities except non-agricultural labour outside the household (i.e. wage or casual labour).

While we find no statistically significant differences between young women and men in terms of exposure to climatic stress, Table 4 shows that young women are characterized by a variety of individual-level features that are associated with increased vulnerability. Young women are 11 percentage points less likely to currently attend school and have almost a third of a year less of education. In addition, 35 percent of young women are married, compared to only 12 percent of young men.

Moreover, young women tend to live in households that are poorer, smaller and younger, have a higher share of dependent members and a lower share of working members. Interestingly, the average education levels in the households where young women live are slightly higher, compared to the households where young men live. However, while all these differences are statistically significant, they are small in magnitude. One striking difference regards the ratio of female over male household members, which is much higher in the households where young women live (1.65 vs 0.91). Thus, young women live in households with more than 1.5 times as many women as men. Young men, in contrast tend to live in households with slightly more male than female members.

Table 3. Descriptive statistics for all working-age individuals, by age group

	(1) Working- age (15 – 64)	(2) Youth (15 – 24)	(3) Adults (25 – 64)	(4) Difference (2) – (3)	N
<i>Labour variables</i>					
Has a job	0.74	0.62	0.81	-0.20***	55 709
Worked in last 7 days	0.65	0.52	0.73	-0.21***	55 709
Hours per week (if worked)	26.46	22.42	28.25	-5.83***	36 319
Worked in agriculture (if worked)	0.85	0.89	0.83	0.06***	36 319
Hours per week (if worked in sector)	21.91	19.67	22.98	-3.31***	30 859
Worked on HH farm (if worked)	0.77	0.81	0.76	0.05***	36 319
Hours per week (if worked in sector)	20.94	18.94	21.88	-2.94***	28 126
Worked in ag. wage or casual labour (if worked)	0.14	0.15	0.14	0.01**	36 319
Hours per week (if worked in sector)	17.21	15.53	18.01	-2.48***	5 069
Worked in non-agriculture (if worked)	0.25	0.16	0.29	-0.13***	36 319
Hours per week (if worked in sector)	31.56	30.01	31.95	-1.94***	9 024
Worked in HH non-farm business (if worked)	0.17	0.10	0.19	-0.09***	36 319
Hours per week (if worked in sector)	28.53	25.94	29.15	-3.20***	6 042
Worked in non-ag. wage or casual labour (if worked)	0.09	0.06	0.10	-0.04***	36 319
Hours per week (if worked in sector)	36.23	36.36	36.19	0.17	3 104
<i>Exposure to climate stressors</i>					
Heat stress: number of days with extreme temperatures in last year	5.30	5.18	5.37	-0.19***	55 709
Extreme precipitation: no. of days with extreme pre. in last year	5.84	6.03	5.72	0.31***	55 709
Drought: exposure to drought in last year (yes/no)	0.19	0.20	0.19	0.02***	55 709
<i>Individual characteristics</i>					
Respondent is female	0.52	0.50	0.54	-0.04***	55 709
Age of the respondent	31.84	18.91	39.90	-20.98***	55 709
Respondent's years of education	5.53	6.62	4.85	1.77***	55 709
Respondent's marital status	0.57	0.23	0.78	-0.55***	55 709
Respondent currently attends school	0.16	0.40	0.01	0.38***	55 709

Table 3. Descriptive statistics for all working-age individuals, by age group

	(1) Working- age (15 – 64)	(2) Youth (15 – 24)	(3) Adults (25 – 64)	(4) Difference (2) – (3)	N
<i>Household characteristics</i>					
Poor HH	0.21	0.17	0.23	-0.05***	55 709
Female HH head	0.20	0.24	0.17	0.06***	55 709
HH size	6.22	6.48	6.05	0.43***	55 709
Avg. age in HH	23.93	23.21	24.37	-1.17***	55 709
Avg. education in HH	4.60	5.04	4.33	0.71***	55 709
Share of dependent HH members (<15 and >64)	0.39	0.34	0.42	-0.07***	55 709
Ratio of female and male HH members	1.28	1.28	1.29	-0.01	55 709
Share of working HH members	0.35	0.41	0.32	0.09***	55 709
HH receives social protection	0.19	0.19	0.19	0.00	55 709
HH engaged in agriculture	0.91	0.91	0.92	-0.01***	55 709
Observations	55 709	21 392	34 317		

Notes: All labour hour variables account for hours worked in the main and secondary job. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Authors' own elaboration.

Table 4. Descriptive statistics for all working-age youth, by gender

	All youth	Young women	Young men	Difference (2) – (3)	N
<i>Labour variables</i>					
Has a job	0.62	0.58	0.65	-0.07***	21 392
Worked in last 7 days	0.52	0.48	0.56	-0.08***	21 392
Hours per week (if worked)	22.42	19.69	24.74	-5.05***	11 186
Worked in agriculture (if worked)	0.89	0.89	0.90	-0.01	11 186
Hours per week (if worked in sector)	19.67	17.06	21.86	-4.81***	9 994
Worked on HH farm (if worked)	0.81	0.82	0.80	0.02***	11 186
Hours per week (if worked in sector)	18.94	16.74	20.85	-4.11***	9 041
Worked in ag. wage or casual labour (if worked)	0.15	0.12	0.17	-0.05***	11 186
Hours per week (if worked in sector)	15.53	12.10	17.60	-5.50***	1 631
Worked in non-agriculture (if worked)	0.16	0.16	0.16	0.00	11 186
Hours per week (if worked in sector)	30.01	27.64	32.06	-4.42***	1 807
Worked in HH non-farm business (if worked)	0.10	0.12	0.09	0.03***	11 186
Hours per week (if worked in sector)	25.94	24.36	27.72	-3.36***	1 168
Worked in non-ag. wage or casual labour (if worked)	0.06	0.05	0.07	-0.03***	11 186
Hours per week (if worked in sector)	36.36	35.05	37.07	-2.02	658
<i>Exposure to climate stressors</i>					
Heat stress: number of days with extreme temperatures in last year	5.18	5.16	5.20	-0.04	21 392
Extreme precipitation: no. of days with extreme pre. in last year	6.03	6.04	6.02	0.02	21 392
Drought: exposure to drought last year (yes/no)	0.20	0.21	0.20	0.00	21 392
<i>Individual characteristics</i>					
Respondent is female	0.50	1.00	0.00	1.00	21 392
Age of the respondent	18.91	19.02	18.81	0.21***	21 392
Respondent's years of education	6.62	6.48	6.77	-0.29***	21 392
Respondent's marital status	0.23	0.35	0.12	0.23***	21 392
Respondent currently attends school	0.40	0.34	0.45	-0.11***	21 392
<i>Household characteristics</i>					
Poor HH	0.17	0.18	0.17	0.01***	21 392
Female HH head	0.24	0.24	0.24	0.00	21 392
HH size	6.48	6.17	6.79	-0.62***	21 392
Avg. age in HH	23.21	22.15	24.26	-2.11***	21 392
Avg. education in HH	5.04	5.13	4.96	0.17***	21 392
Share of dependent HH members (<15 and >64)	0.34	0.35	0.33	0.02***	21 392
Ratio of female and male HH members	1.28	1.65	0.91	0.74***	21 392
Share of working HH members	0.41	0.40	0.41	-0.01***	21 392
HH receives social protection	0.19	0.18	0.19	-0.01**	21 392
HH engaged in agriculture	0.91	0.90	0.92	-0.02***	21 392
Observations	21 392	10 661	10 731		

Notes: All labour hour variables account for hours worked in the main and secondary job. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Authors' own elaboration

Empirical results

Younger and older people's likelihood to work and weekly working times

Table 5 shows the relationship between exposure to the three climate stressors and total weekly working times for older people (coefficient β_2) and the differential effect on youth (coefficient β_3 of the interaction term). In addition, we calculated the sum of the two former coefficients, which shows the average net effect of the climate stressor on youth. All results are transformed into average marginal effects so that they can be interpreted in percentage points for the first step and in minutes for the second step.

The results in the first column of Table 5 show a positive association between exposure to heat stress and young people's likelihood of working. Specifically, with each day of extreme temperatures, young people are about 0.1 percentage points more likely to work. For older people, in contrast, this relationship is negative and equal to about -0.3 percentage points per day. Moreover, working individuals increase their weekly working hours in times of heat stress. For older people, this effect corresponds to about 12 minutes and for youth to about 22 minutes for each day of extreme heat.

Table 5. Effects of climate stress on the likelihood to work and weekly working hours of younger and older working-age individuals (conditional average marginal effects)

	(1) Heat stress	(2) Extreme precipitation	(3) Droughts
<i>First step (probit)</i>			
Youth	-0.0673*** (0.0093)	-0.0190** (0.0094)	-0.0474*** (0.0089)
Climate stressor	-0.0029*** (0.0005)	0.0036*** (0.0007)	0.0008 (0.0066)
Youth x Climate stressor	0.0041*** (0.0007)	-0.0046*** (0.0007)	0.0087 (0.0095)
Average net effect of climate stressor on youth	0.0011* (0.0006)	-0.0009 (0.0006)	0.0095 (0.0075)
<i>Second step (OLS)</i>			
Youth	-1.5e+02*** (38.4380)	-1.0e+02 (38.8699)	-75.3860** (34.1404)
Climate stressor	12.4704*** (2.0880)	-5.7658*** (2.0147)	62.3107*** (24.0975)
Youth x Climate stressor	9.8284*** (3.3983)	-0.6754 (2.6003)	-134.7908*** (38.8900)
Average net effect of climate stressor on youth	22.2988*** (2.9984)	-6.4412*** (2.3746)	-72.4801** (34.5235)
N first step		55 709	
N second step		36 319	
Mean outcome second step (hours)		26.46	

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Extreme precipitation has the opposite effects compared to heat stress. Specifically, one day of exposure is associated with a substantial increase in older people's likelihood of working by about 0.4 percentage points. In contrast, young people's likelihood to work does not change under exposure to extreme precipitation. Moreover, the results point to a decrease in weekly working hours among the working population. For both older and younger people, the effect amounts to about six minutes per day of exposure.

We do not find a statistically significant relationship between exposure to droughts and the likelihood of working. However, droughts are associated with an increase in the weekly working hours of older people by about one hour. For young people, in contrast, the results point to a decrease in working hours by slightly more than one hour. These effects reflect yearly exposure to droughts.

Young women's and men's likelihood to work and weekly working times

Table 6 presents the results of total weekly working times in the sub-sample of youth, with an interaction term for female gender. When exposed to one additional day of heat stress, young women's likelihood to work reduces by 0.6 percentage points, while men are 0.3 percentage points more likely to work. Thus, the generally higher propensity to work among youth is driven by young men. In addition, young women work about 13 minutes less per week compared to young men for each day of extreme heat.

Similar findings from South Africa have been at least partially explained by men's higher resistance to heat, which stems from physiological differences in thermoregulation between women and men (Shayegh and Dasgupta, 2024). Moreover, young women's reduced labour participation may be linked to an increase in their unpaid work burden, e.g. in terms of fetching water or other household chores. Young women are also more involved in caring for children and elderly dependents, whose health is relatively more susceptible to heat (Foster *et al.*, 2020; Helldén *et al.*, 2021; Larose *et al.*, 2013). However, the limited evidence that exists on the relationship between extreme weather events and women's unpaid care burden is mixed. A quantitative study from China indicates that women spend less time on chores and childcare when exposed to high temperatures. In contrast, albeit not climate-related, a study from El Salvador found that female household members spent significantly more hours doing household chores and care work after an earthquake (Halliday, 2012).

In contrast to heat stress, exposure to extreme precipitation is associated with an increase in young women's likelihood of working relative to that of young men by 0.5 percentage points per day of extreme precipitation. As mentioned earlier, case studies from Madagascar, Colombia, Guatemala and Mexico find a general increase in the likelihood to be employed for adolescents and youth, though with a stronger effect for males than for females observed in the three Latin American countries (Acevedo, 2016; Baez *et al.*, 2017; de Janvry *et al.*, 2006; Marchetta, Sahn and Tiberti, 2019). Moreover, for each day of extreme precipitation, working young women dedicate 9 minutes more per week to their jobs than young men. Thus, while young people generally tend to work less when exposed to extreme precipitation, the opposite is the case for young women.

Similarly, droughts are associated with an increased gap in young women's and men's likelihood to work by about 3 percentage points per year. The study from Madagascar produced similar results (Marchetta, Sahn and Tiberti, 2019). In contrast, research from Mexico and India has found general decreases in children's and young people's likelihood of employment, with larger effects among females compared to males detected in Mexico (de Janvry *et al.*, 2006; Shah and Steinberg, 2017).

Regarding both extreme precipitation and droughts, the discrepancies between our findings and previous studies relying on country-level data from India and Latin America might arise due to regional discrepancies in gender norms. In fact, women's role in the labour market is more pronounced in SSA than in other regions of the world. Female labour force participation stands at 62 percent in SSA, compared to 29 percent in South Asia and 57 percent in Latin America (World Bank Data, 2022). Relative to South Asia, acceptance of women working outside the household is significantly higher in SSA, while it is comparable between SSA and Latin America (ILO, 2017). However, rural women in Latin America face higher rates of care-related inactivity compared to rural women in SSA, who are more likely to engage in informal employment to balance caregiving and paid work (ILO, 2024c).

Table 6. Effects of climate stress on the likelihood to work and weekly working hours of young women and men (conditional average marginal effects)

	(1) Heat stress	(2) Extreme precipitation	(3) Droughts
<i>First step (probit)</i>			
Female	-0.0834*** (0.0089)	-0.1625*** (0.0092)	-0.1360*** (0.0074)
Climate stressor	0.0028*** (0.0009)	0.0016 (0.0010)	-0.0161 (0.0116)
Female x Climate stressor	-0.0088*** (0.0012)	0.0053*** (0.0011)	0.0311** (0.0146)
Average net effect of climate stressor on young women	-0.0060*** (0.0009)	0.0070*** (0.0010)	0.0150 (0.0112)
<i>Second step (OLS)</i>			
Female	-309.7258*** (34.4214)	-415.6936*** (37.5201)	-368.9273*** (27.8682)
Climate stressor	18.4372*** (3.5108)	-2.9262 (3.1196)	2.5240 (39.7044)
Female x Climate stressor	-12.5940** (4.9028)	9.4814*** (3.5264)	-26.8477 (53.4003)
Average net effect of climate stressor on young women	5.8432 (3.8764)	6.5551** (3.2712)	-24.3238 (43.1125)
N first step		21 392	
N second step		11 186	
Mean outcome second step (hours)		22.42	

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own calculations.

Sectoral shifts of workers under exposure to the different climate stressors

Table 7 to Table 9 disentangle how working individuals move between agricultural and non-agricultural activities and adjust their working times in these sectors in the face of extreme weather events. The first and third columns refer to the full sample, with an interaction term for young age, while the second and fourth columns refer to the sub-sample of youth, with an interaction term for female gender. All models are restricted to those individuals who worked at least one hour in the week preceding the interview.

Heat stress

Rural workers of all ages are less likely to engage in agriculture when exposed to extreme heat (Table 7). However, among youth, significant gender differences prevail. Each day of heat stress decreases young women's likelihood to engage in agriculture by 0.6 percentage points, compared to 0.3 percentage points for young men. Moreover, young men who remain in agriculture increase the time spent on farm by about 12 minutes per week for each day of exposure to heat. In contrast, young women's working time does not change. Thus, young men who keep engaging in agriculture compensate for the decreased participation in this type of work by older people and young women by working more. Biological factors that make young men more resistant to heat compared to both older people and women of the same age might explain this result (Foster *et al.*, 2020; ILO, 2024b; Larose *et al.*, 2013).

These results partly correspond to findings from rural China, which show that female farmers reduce their working times when exposed to heat stress. Yet, the study also finds a reduction of working times among male farmers, though the effect is weaker than among women (Garg, Gibson and Sun, 2020). In contrast to our findings, a previous cross-country study finds that adult women from rural areas in 31 African countries decrease their working times in agriculture in response to heat stress, but to a smaller extent than men (Nico and Azzarri, 2024). Similar evidence is produced in a country-level study from the United Republic of Tanzania (Lee *et al.*, 2021). However, the comparability of our results with these three studies is limited due to differences in the age cohorts included in the sample, the definition of climate shocks, the precision of the climate data, and the definition of outcome variables.

The heat-related exodus of labour from the agricultural sector is partly compensated for by increased participation in non-agricultural work. Rural workers of all ages are more likely to engage in the non-agricultural sector after a heat shock. This effect is strongest for young women, amounting to 0.8 percentage points compared to 0.5 percentage points among their male counterparts. A differentiation of these effects into labour in and outside the household (see Appendix A3) reveals that the difference between young women and men arises due to young women's increased participation in non-agricultural wage or casual employment outside the household. Moreover, with each day of heat stress, young women and men who engage in non-agricultural activities increase their weekly working times by 22 and 40 minutes, respectively.

Thus, overall, exposure to heat stress is associated with a shift of rural workers from the agricultural to the non-agricultural sector. Among youth, this shift is more pronounced for women. However, the welfare implications of this trend depend on the quality of the jobs

that are offered outside of agriculture for young people, especially for young women. Moreover, as described earlier, young people's level of engagement in non-agriculture remains low, hovering around 16 percent of working youth.

Table 7. Effects of heat stress on younger vs older and young female vs young male workers' likelihood to work and weekly working hours in agriculture and non-agriculture (conditional average marginal effects)

	Agriculture		Non-agriculture	
	(1) Youth vs adults	(2) Young women vs men	(3) Youth vs adults	(4) Young women vs men
<i>First step (probit)</i>				
Sub-group	0.0235*** (0.0089)	0.0095 (0.0087)	-0.0387*** (0.0105)	-0.0205** (0.0102)
Heat stress	-0.0049*** (0.0004)	-0.0031*** (0.0008)	0.0064*** (0.0005)	0.0045*** (0.0010)
Sub-group x Heat stress	0.0001 (0.0008)	-0.0032*** (0.0011)	0.0015 (0.0009)	0.0037*** (0.0013)
Average net effect of heat stress on sub-group	-0.0048*** (0.0007)	-0.0063*** (0.0008)	0.0078*** (0.0008)	0.0082*** (0.0010)
<i>Second step (OLS)</i>				
Sub-group	-57.8750* (33.4437)	-261.3814*** (30.7819)	-1.2e+02 (90.4568)	-160.1225 (124.7119)
Heat stress	1.2852 (1.9335)	12.2839*** (3.2046)	26.8674*** (4.4489)	40.4704*** (12.5013)
Sub-group x Heat stress	10.4799*** (2.9929)	-14.8299*** (4.4371)	-15.1484* (9.3843)	-18.4714 (16.3497)
Average net effect of heat stress on sub-group	11.7651*** (2.6650)	-2.5460 (3.5500)	11.7190 (8.8185)	21.9990* (12.8466)
N first step	36 319	11 186	36 319	11 186
N second step	30 859	9 994	9 024	1 807
Mean outcome second step (hours)	21.91	19.67	31.56	30.01

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Extreme precipitation

When extreme precipitation occurs, we observe a similar shift of rural workers from agriculture to non-agriculture as during heat stress, though with generally smaller effect sizes and different gender dynamics among youth (Table 8). One additional day of exposure reduces working youths' likelihood to engage in agriculture by 0.2 percentage points, while increasing their likelihood to engage in non-agricultural activities by 0.3 percentage points. However, this effect is driven entirely by young men, while young women's labour allocations between the two sectors do not change. Moreover, extreme precipitation is associated with a reduction in young men's working time dedicated to agriculture by 8 minutes per day of exposure, while no such effect is detectable for young women. Among older adults, we also observe a shift from agricultural to non-agricultural activities, though with smaller effect sizes compared to youth.

Thus, when extreme precipitation occurs, young women do not follow the general trend of decreased participation in agriculture and take on a higher labour burden in farming compared to young men. A similar effect is found by Moshoeshe, Nchake and Tiberti (2020) in Lesotho, where adult men were less likely to work in agriculture after a flood, while women's labour allocations remained unchanged. In contrast, Nico and Azzarri (2024) find the opposite trend in their study of 31 African countries. However, as indicated earlier, the comparability of their study with ours is limited.

Appendix A3 differentiates these broad sectorial shifts by household-level labour and employment outside the household. In terms of non-agricultural activities, the increase in young people's likelihood to work in a non-farm household business is stronger than the increase in their likelihood of engaging in non-agricultural wage or casual employment. Moreover, youth who engage in non-agricultural wage or casual employment significantly increase the time they dedicate to these activities by about an hour for each day of exposure. For adults, in contrast, the effect on working times in the sector is negative. This age and gender pattern probably reflects the type of work that is available after extreme precipitation events, which can cause substantial damage to physical infrastructure. Specifically, repair work and recovery efforts involve physically straining work that is better suited for younger than for older people, and deemed more appropriate for men compared to women.

As for heat stress, the welfare implications of these developments depend on the kind of wage or casual employment that young people are engaged in. If the types of jobs available after extreme precipitation events are mainly in reconstruction of damaged facilities and infrastructure, they are likely to be on a short-term and piece-meal basis, rather than formalized opportunities for skill development and salary growth.

Table 8. Effects of extreme precipitation on younger vs older and young female vs young male workers' likelihood to work and weekly working hours in agriculture and non-agriculture (conditional average marginal effects)

	Agriculture		Non-agriculture	
	(1) Youth vs adults	(2) Young women vs men	(3) Youth vs adults	(4) Young women vs men
<i>First step (probit)</i>				
Sub-group	0.0324*** (0.0089)	-0.0297*** (0.0086)	-0.0351*** (0.0112)	0.0228** (0.0105)
Extreme precipitation	-0.0009* (0.0005)	-0.0022*** (0.0008)	0.0027*** (0.0006)	0.0043*** (0.0010)
Sub-group x Extreme precipitation	-0.0012* (0.0007)	0.0034*** (0.0009)	0.0005 (0.0008)	-0.0037*** (0.0011)
Average net effect of extreme precipitation on sub-group	-0.0021*** (0.0006)	0.0012 (0.0008)	0.0032*** (0.0008)	0.0006 (0.0009)
<i>Second step (OLS)</i>				
Sub-group	-5.2472 (34.2590)	-415.6936*** (32.9841)	-4.2e+02*** (98.2700)	-258.1487** (123.0439)
Extreme precipitation	-8.6667*** (1.7842)	-8.2973*** (2.8583)	-20.9970*** (5.0048)	-3.4093 (10.4102)
Sub-group x Extreme precipitation	-0.3354 (2.3278)	12.8073*** (3.2848)	30.9960*** (7.4456)	-1.2181 (12.1921)
Average net effect of extreme precipitation on sub-group	-9.0021*** (2.2372)	4.5101 (2.8454)	9.9990 (7.3931)	-4.6273 (11.3767)
N first step	36 319	11 186	36 319	11 186
N second step	30 859	9 994	9 024	1 807
Mean outcome second step (hours)	21.91	19.67	31.56	30.01

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Droughts

If a drought occurs, rural workers shift from non-agriculture to agriculture, i.e. in the opposite direction compared to the two other shocks (Table 9). In a year with droughts, both younger and older workers are around 2 percentage points more likely to engage in agriculture. Moreover, older workers' likelihood of working in the non-agricultural sector decreases by about 2 percentage points, while young people's participation in this sector remains unaffected. Among youth, the increased participation in agriculture is driven by young women, and as shown in the supplementary results in Appendix A3, by work on the household farm, while participation in agricultural wage or casual employment tends to decrease. This effect likely reflects the decreased availability of agricultural jobs during droughts. However, among youth, the effect is only statistically significant for men, while young women's participation remains unaffected. This finding underscores the important role of young women's work in agriculture during droughts.

Table 9. Effects of droughts on younger vs older and young female vs young male workers' likelihood to work and weekly working hours in agriculture and non-agriculture (conditional average marginal effects)

	Agriculture		Non-agriculture	
	(1) Youth vs adults	(2) Young women vs men	(3) Youth vs adults	(4) Young women vs men
<i>First step (probit)</i>				
Sub-group	0.0246*** (0.0082)	-0.0125* (0.0066)	-0.0347*** (0.0101)	0.0036 (0.0083)
Drought	0.0230*** (0.0059)	-0.0015 (0.0105)	-0.0199*** (0.0068)	0.0093 (0.0121)
Sub-group x Drought	0.0005 (0.0109)	0.0326** (0.0149)	0.0122 (0.0125)	-0.0294* (0.0173)
Average net effect of droughts on sub-group	0.0235** (0.0094)	0.0310** (0.0121)	-0.0077 (0.0109)	-0.0201 (0.0134)
<i>Second step (OLS)</i>				
Sub-group	23.7842 (31.5214)	-331.0064*** (25.9063)	-122.9333 (81.0608)	-251.5701** (98.8292)
Drought	81.3772*** (21.3037)	9.7700 (35.8135)	46.2551 (62.4110)	-1.3e+02 (160.4578)
Sub-group x Drought	-131.2252*** (34.6478)	-17.9008 (49.8974)	-489.7143*** (130.3080)	-1.1e+02 (228.6149)
Average net effect of droughts on sub-group	-49.8480* (29.4739)	-8.1308 (39.9474)	-443.4592*** (114.9940)	-231.9283 (183.1749)
N first step	36 319	11 186	36 319	11 186
N second step	30 859	9 994	9 024	1 807
Mean outcome second step (hours)	21.91	19.67	31.56	30.01

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

At the same time, as under exposure to extreme precipitation, young women are relatively less likely than young men to engage in the non-agricultural sector when droughts occur. Decreased participation of women in non-agricultural jobs due to droughts is also found

among youth in India (Afridi, Mahajan and Sangwan, 2022) and among youth and adults in Lesotho (Moshoeshoe, Nchake and Tiberti, 2021). The study from India links this result to women's mobility restrictions, while according to the study from Lesotho, this trend arises because most of women's non-agricultural jobs are linked to agriculture, e.g. sales of grains or livestock, and therefore more susceptible to droughts (Moshoeshoe, Nchake and Tiberti, 2021). In addition, the finding likely reflects women's responsibility in procuring food for the household, which becomes more time-consuming as agricultural productivity contracts in times of drought. On the other hand, both studies also find that men are more likely to work in non-agricultural activities, which is not supported by our findings.

Discussion and conclusions

In the face of climate stressors, young people in rural SSA adjust their labour outcomes in different ways than older people do, with important distinctions between young women and young men. In summary, our analysis reveals two key findings, which demonstrate that policies aimed at addressing the relationship between climate and labour must account for age and gender-related heterogeneities.

First, the analysis uncovers some age-dependent differences in the impact of climate stressors on rural people's labour outcomes. Under heat stress, the rural societies represented by our sample rely more on the labour of youth, particularly of young men. This finding is likely driven by biological factors, which make young men more resistant to heat compared to older people and women of the same age. The welfare implications of this trend depend on whether young men enter the labour market at the expense of education, as well as on the type of job they do. In contrast, when exposed to extreme precipitation and droughts, older adults contribute relatively more labour than young people.

These results underscore the importance of treating youth as a unique group, which faces challenges and opportunities that differ from the rest of population. Policies that seek to buffer the adverse effects of extreme weather events should therefore be age sensitive. At present, such age sensitivity is sorely lacking. For example, FAO (2024) analyzed the nationally-determined contributions (NDCs) and national adaptation plans (NAPs) of 24 countries and found that only 2 percent of the climate actions explicitly refer to youth.

Our second key finding is that exposure to climate stressors is associated with widening gender gaps in young people's labour outcomes, mostly to the detriment of young women. For example, exposure to heat stress widens the gender gap in the overall likelihood of engaging in productive labour by about 0.9 percentage points per day of exposure. In the full sample of youth, i.e. regardless of exposure to heat stress, 56 percent of young men worked in the week preceding the interview, compared to 48 percent of young women (see Table 4 of descriptive statistics). Thus, each day of heat stress widens the existing gap by more than 10 percent. The widening of this gap means that young women miss out on valuable opportunities to develop marketable skills and relevant working experience, thereby falling further behind young men in the labour market.

In contrast, exposure to extreme precipitation and droughts narrows the gap between young women's and men's overall likelihood to work but induces a concentration of young female workers in the agricultural sector and of young male workers in the non-agricultural sector.

In the full sample, we do not observe gender gaps in terms of young workers' participation in the agricultural and non-agricultural sector (Table 4). However, under average exposure to extreme precipitation, the participation rates begin to drift apart by about 0.4 percentage points per day of exposure, and a year with a drought is associated with a gap of 3 percentage points. While these results seem small, it is important to acknowledge that they represent effects of only one day or year of exposure to extreme weather, and that these events are set to become increasingly frequent. Moreover, given that agricultural labour tends to be less remunerative than non-agricultural labour (FAO, 2024), this development is likely to also contribute to widening gender wage gaps.

Our findings are critical in the context of rural SSA, where a lack of economic opportunities has led to rampant youth unemployment and underemployment rates (Sumberg *et al.*, 2021). Protecting and expanding young people's labour opportunities in the face of climate change will be critical for countries in SSA, as they seek to capitalize on their youth dividends to spur economic growth. To this end, climate and labour policies must account for gender and age-related differences. Social protection policies can be strong instruments to this end. For example, in India, the public works programme of the National Rural Employment Guarantee Scheme (NREGS) provided working opportunities for adult men and women affected by flooding (Maitra and Tagat, 2024). In SSA, however, less than 30 percent of the population exposed to extreme weather events is covered by social protection schemes (World Bank, 2024), and gender- and youth-sensitive programming is rare (Devonald *et al.*, 2024). In addition, skills training programs and entrepreneurship promotion interventions can equip young people with opportunities to diversify their income sources (Kluve *et al.*, 2019). At the same time, policymakers in SSA need to tackle the lack of wage job opportunities which lies at the core of the continent's youth employment issues (Sumberg *et al.*, 2021). Lastly, continued work is needed to dismantle social norms that trap women in disadvantaged labour market positions. Addressing these structural factors will require concerted efforts at the national and global levels.

This study has several limitations. As indicated earlier, we cannot make rigorous claims about the causal relationship between exposure to climate stress and labour outcomes, given that our data is cross-sectional. Other weaknesses are mainly related to insufficient availability of data. Our analysis does not account for losses or gains in labour productivity and wages. Therefore, we cannot make statements about the welfare implications of the identified relationships between climate stress and labour outcomes. Furthermore, as we only account for individuals' main and secondary jobs, our analyses of engagement in agriculture and non-agriculture and of weekly working times provide an incomplete perspective of the livelihoods of rural youth, which may entail more than two income-generating activities. In addition, by focusing on productive labour, we do not capture changes in time allocation to

reproductive household and care work, which take up a significant share of young women's days. It would be particularly interesting to examine substitution effects between productive and reproductive work in the face of climatic stress. Moreover, as we have no measure of prevailing gender norms, we can only assume that these norms partly drive the detected effects. Lastly, our results show the effect of climate stressors on people who live in rural areas after exposure. Thus, we cannot verify whether rural people, particularly young men, migrate to urban areas in response to climate stressors, and whether this mobility mediates the labour outcomes of those who stay behind.

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Appendices

Variable construction

Table A1. provides an overview and comparison of how the outcome variables (labour times and sectors) are defined in the sampled countries. The sub-sections below provide further details.

Table A 1. Construction of outcome variables

Country	Identification of main/secondary job	Recall period working hours	Recall period sectors of wage/casual labour
Ethiopia	Ranking of hours worked in 4 different activities	Last 7 days	Wage and casual: last 12 months
Malawi	Ranking of hours worked in 5 different activities	Last 7 days	Wage: last 12 months Casual: last 7 days
Niger	Main and secondary job identified by respondent	Usual hours in last 12 months	Wage and casual: last 7 days
Nigeria	Ranking of hours worked in 3 different activities	Last 7 days	Wage and casual: last 7 days
United Republic of Tanzania	Ranking of hours worked in 5 different activities	Last 7 days for HH agriculture, wage or casual labour, and helping in a non-farm HH business; usual hours in last 12 months for running a non-farm business	Wage and casual: last 7 days
Uganda	Main and secondary job identified by respondent	Last 7 days	Wage and casual: last 7 days

Source: authors' own elaboration

Ethiopia

The questionnaire asks for the hours worked in the *last 7 days* in the following activities:

- (1) household agricultural activities (including livestock and fishing-related activities),
- (2) run or help with any kind of non-agricultural or non-fishing household business, big or small, for his or herself or for the household,
- (3) casual, part-time, or temporary labour, and
- (4) work for a wage, salary, commission, or any payment in kind, excluding temporary.

Main and secondary jobs are determined by ranking the hours worked in (1) to (4) and selecting the top two. 61 observations do not have a unique first or second rank and are dropped from the analysis. Unpaid apprentices are excluded.

To identify the sectors of (3) and (4), we use information from the module on the respondent's main job, defined as any kind of job, including part-time labour, for wage, salary, commission or any payment in kind, for anyone who is not a member of the household (excluding temporary work), over the *last 12 months*. The following question is used, regardless of whether (3) and (4) are the main or the secondary job:

What is the main activity of this business or organization where [NAME] works?

[Answer options: Agriculture, mining, manufacturing, etc.]

The module on temporary work does not elicit the sector of operations. However, the information from the wage labour module allows us to classify all relevant observations.

Malawi

The questionnaire asks for the hours worked in the *last 7 days* in the following activities:

- (1) on household farming activities whether for sale or for household food,
- (2) household livestock activities whether for sale or for household food,
- (3) household fishing activities whether for sale or for household food,
- (4) run or do any kind of non-agricultural or non-fishing household business, big or small, for yourself,
- (5) help in any of the household's non-agricultural or non-fishing household businesses,

(6) casual, part-time or ganyu labour, and

(7) any work for a wage, salary, commission, or any payment in kind, excluding ganyu.

(1) to (3) are aggregated. Subsequently, main and secondary jobs are determined by ranking the hours worked in (1)+(2)+(3) to (7) and selecting the top two. 427 do not have a unique first or second rank and are dropped from the analysis. Unpaid apprentices are excluded.

We use the following questions to determine the sector of (6):

- a) In what type of economic activity did you spend most of your time in the *last 7 days*? Provide main activity.

[Answer options: wage employment excluding ganyu, household business (non- agricultural), unpaid household labour (agricultural), unpaid apprenticeship, ganyu]

- b) In what type of economic activity did you spend most of your time in the *last 7 days*? Provide secondary activity.

[Answer options: as a)]

If most hours are worked in (6), they are classified as agricultural if the answer to a) or b) is “ganyu”, and as non-agricultural otherwise. The same rule is applied if second-most hours are worked in (6).

We use the following question to determine the sector of (7):

Describe what kind of trade or business your main wage job over the *last 12 months* is connected with.

[Answer options: Agriculture (with sub-sectors), mining (with sub-sectors), manufacturing (with sub-sectors), etc.]

If most hours are worked in (7), they are classified as agricultural if the is “agriculture”, and as non-agricultural if the answer is not “agriculture”. The same rule is applied if second-most hours are worked in (7).

Niger

The labour module is split into two parts: the first about the main job and the second about the secondary job that respondents have had in the *last 7 days*. Both are asked to all respondents who worked for at least one hour in one of the following activities: agricultural activities (with or without pay, on the household’s own farm or on someone else’s farm); in a business on one’s own behalf or on behalf of another member of the household (with or without pay); for a company, for the government, for an employer, or for any other person who is not a member of your household; and as an apprentice (with or without pay).

Hours worked in the main and secondary job were determined by taking the product of the following questions posed within both sub-modules:

- (1) How many days per week on average does [NAME] *usually* devote to this job?
- (2) How many hours per day on average does [NAME] *usually* spend on this job?

The sectors of the main and secondary jobs were determined through the following question, posed within both sub-modules:

What is the business activity of the company where [NAME] worked, or what products (services) does it manufacture (provide)?

[Answer options: Agriculture (with sub-sectors), mining (with sub-sectors), manufacturing (with sub-sectors), etc.]

Unpaid apprentices are excluded.

Nigeria

The questionnaire asks for the hours worked, in the *last 7 days*, in the following activities:

- (1) worked for someone who is not a member of your household, for example, an enterprise, company, the government or any other individual for payment in cash or in-kind.
- (2) worked on a farm owned or rented by this household, either in cultivating crops or in other farming tasks; cared for livestock belonging to this household; or gone fishing or worked in fish farming owned by the household, and
- (3) worked on their own account or in a business enterprise belonging to [NAME] or another household member, for example, as a trader, shopkeeper, barber, dressmaker, carpenter or taxi driver.

Main and secondary jobs are determined by ranking the hours worked in (1) to (3) and selecting the top two. 57 observations do not have a unique first or second rank and are dropped from the analysis. Unpaid apprentices are excluded.

To identify the sector of (1), we use information from the module on the respondent's main wage/salaried job, over the *last 7 days*, which is asked to all respondents whose answer to (1) is at least 1 hour. The following question is used:

In what sector is [NAME]'s main activity?

[Answer options: Agriculture, mining, manufacturing, etc.]

United Republic of Tanzania

The questionnaire asks whether, in the *last 7 days*, respondents worked for at least one hour in the following activities:

- (1) employee for a wage, salary, commission or any payment in kind; including doing paid apprenticeship, domestic work or paid farm work,
- (2) agricultural activities (including farming, raising livestock or fishing, whether for sale or for household food),
- (3) run any kind of non-farm business of any size for themselves or the household, and
- (4) help in any kind of non-farm business of any size for themselves or the household.

If the answer to (1) is “yes” the questionnaire asks for the hours worked in a main and secondary wage job in the *last 7 days* (1a and 1b).

If the answer to (2) or (3) is yes, the questionnaire asks for the hours worked in the *last 7 days* in the respective activity.

Regarding (4), there is no question about the hours worked in the last 7 days in *running* a non-farm business. However, if the answer to (3) or (4) is “yes” respondents are asked for the usual weekly hours worked in the *last 12 months*. Thus, if the respondent ran a household business (i.e. (4) = “yes”), we use the hours worked in the *last 12 months* as weekly hours for this activity.

Main and secondary jobs are determined by ranking the hours worked in (1a) to (4) and selecting the top two. 110 observations do not have a unique first or second rank and are dropped from the analysis. Unpaid apprentices are excluded.

To determine the sectors of (1a) and (1b), we use the following question, posed within the sub-modules on both the main and secondary wage job:

What kind of trade or business is it connected with?

[Answer options: Agriculture (with sub-sectors), mining (with sub-sectors), manufacturing (with sub-sectors), etc.]

Uganda

The questionnaire asks whether in the *last 7 days* respondents worked for at least one hour in the following activities:

- (1) work for a wage, salary, commission or any payment in kind, from work in agriculture or non-agriculture, and including doing paid domestic work,
- (2) run a business of any size, for themselves or another household member,
- (3) help without being paid in any kind of business run by this household,
- (4) an apprentice, and
- (5) work on this household's farm.

If the answer to any of the above is “yes”, sub-modules about the main and secondary job are elicited, asking for both hours worked in the *last 7 days* and the sector of the activity. Unpaid apprentices are excluded.

Full regression outputs (coefficients)

Table A 2. Effects of heat stress on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Heat stress (1 yr)	-0.009*** (0.002)	-0.026*** (0.003)	0.024*** (0.002)
Young	-0.218*** (0.030)	0.126*** (0.048)	-0.146*** (0.041)
Young # heat stress (1 yr)	0.013*** (0.003)	0.000 (0.004)	0.005 (0.004)
Female	-0.426*** (0.013)	0.069*** (0.019)	-0.141*** (0.016)
Age of the respondent	0.055*** (0.005)	-0.030*** (0.007)	0.059*** (0.006)
Age of the respondent squared	-0.001*** (0.000)	0.000*** (0.000)	-0.001*** (0.000)
Respondent's years of education	0.013*** (0.002)	-0.031*** (0.003)	0.037*** (0.003)
Respondent's marital status	0.044** (0.019)	0.005 (0.029)	0.012 (0.025)
Respondent currently attends school	-0.582*** (0.022)	0.299*** (0.048)	-0.269*** (0.041)
Poor HH	0.039** (0.017)	0.174*** (0.034)	-0.181*** (0.029)
Female HH head	0.088*** (0.018)	-0.010 (0.033)	0.023 (0.029)
HH size	-0.019*** (0.003)	0.001 (0.004)	-0.009** (0.004)
Avg. age in HH	-0.058*** (0.003)	-0.021*** (0.006)	-0.007 (0.005)
Avg. age in HH sq.	0.001*** (0.000)	0.000*** (0.000)	0.000 (0.000)

Table A 2. Effects of heat stress on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Avg. education in HH	-0.014*** (0.004)	-0.053*** (0.006)	0.053*** (0.005)
Share of dependent HH members (<15 and >64)	0.183*** (0.042)	0.013 (0.074)	-0.057 (0.064)
Females/males HH members	0.010* (0.006)	-0.022* (0.012)	0.034*** (0.010)
Share of working HH members	1.557*** (0.031)	0.822*** (0.045)	-0.438*** (0.037)
HH receives social protection	0.055*** (0.016)	0.107*** (0.030)	-0.033 (0.025)
Agricultural household	0.203*** (0.021)	0.824*** (0.037)	-0.573*** (0.036)
URCA	-0.005*** (0.002)	0.003 (0.004)	-0.007** (0.003)
Population density (log)	-0.038*** (0.008)	-0.048*** (0.015)	0.052*** (0.013)
Night-time lights (log)	-0.116*** (0.027)	-0.412*** (0.056)	0.289*** (0.052)
Long-term avg. temp.	-0.022*** (0.003)	-0.010* (0.006)	0.012** (0.005)
Long-term temp. CV	-2.188*** (0.409)	-4.463*** (0.772)	1.791*** (0.655)
Long-term avg. pre.	0.003 (0.004)	0.029*** (0.008)	-0.025*** (0.007)
Long-term pre. CV	-0.237*** (0.040)	-0.361*** (0.077)	0.357*** (0.068)
Malawi	0.254*** (0.024)	0.230*** (0.051)	-0.148*** (0.044)
Niger	1.098*** (0.054)	0.086 (0.100)	0.306*** (0.083)

Table A 2. Effects of heat stress on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Nigeria	0.227*** (0.032)	-0.706*** (0.064)	0.709*** (0.056)
United Republic of Tanzania	0.140*** (0.026)	-0.357*** (0.052)	0.412*** (0.045)
Uganda	0.363*** (0.032)	-0.750*** (0.062)	0.802*** (0.054)
Constant	0.972*** (0.152)	2.572*** (0.278)	-2.507*** (0.238)
N first step	55709	36319	36319

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 3. Effects of heat stress on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Heat stress (1 yr)	0.008*** (0.001)	0.001 (0.002)	0.014*** (0.002)
Young	-0.092*** (0.023)	-0.043* (0.026)	-0.064 (0.048)
Young # heat stress (1 yr)	0.006*** (0.002)	0.008*** (0.002)	-0.008 (0.005)
Female	-0.286*** (0.009)	-0.240*** (0.009)	-0.224*** (0.020)
Age of the respondent	0.029*** (0.003)	0.021*** (0.004)	-0.001 (0.007)
Age of the respondent squared	-0.000*** (0.000)	-0.000*** (0.000)	0.000 (0.000)
Respondent's years of education	0.011*** (0.002)	-0.004** (0.002)	0.018*** (0.003)
Respondent's marital status	-0.015 (0.015)	-0.020 (0.016)	-0.030 (0.028)
Respondent currently attends school	-0.544*** (0.022)	-0.488*** (0.023)	-0.519*** (0.055)
Poor HH	-0.010 (0.016)	0.008 (0.018)	-0.059* (0.032)
Female HH head	0.002 (0.017)	0.004 (0.019)	-0.041 (0.032)
HH size	-0.003 (0.002)	-0.000 (0.003)	-0.002 (0.004)
Avg. age in HH	-0.013*** (0.003)	-0.013*** (0.003)	0.004 (0.006)
Avg. age in HH sq.	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)
Avg. education in HH	0.003 (0.003)	-0.017*** (0.004)	0.013** (0.005)

Table A 3. Effects of heat stress on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	-0.079** (0.039)	-0.076* (0.043)	-0.095 (0.076)
Females/males HH members	0.011* (0.006)	-0.001 (0.007)	0.012 (0.009)
Share of working HH members	0.124*** (0.022)	0.175*** (0.024)	-0.079** (0.039)
HH receives social protection	0.023 (0.016)	0.025 (0.017)	-0.012 (0.029)
Agricultural household	-0.172*** (0.022)	-0.070** (0.031)	-0.198*** (0.029)
URCA	-0.005*** (0.002)	-0.003 (0.002)	-0.004 (0.004)
Population density (log)	-0.015** (0.008)	-0.036*** (0.009)	-0.003 (0.013)
Night-time lights (log)	0.192*** (0.029)	0.230*** (0.042)	0.085** (0.038)
Long-term avg. temp.	-0.004 (0.003)	-0.009*** (0.003)	0.000 (0.006)
Long-term temp. CV	-0.376 (0.385)	-0.286 (0.424)	1.462** (0.736)
Long-term avg. pre.	-0.004 (0.004)	-0.002 (0.004)	0.009 (0.008)
Long-term pre. CV	0.162*** (0.042)	0.169*** (0.047)	0.040 (0.075)
Malawi	-0.471*** (0.027)	-0.453*** (0.030)	-0.183*** (0.060)
Niger	0.489*** (0.048)	0.373*** (0.053)	0.388*** (0.098)
Nigeria	-0.008 (0.034)	-0.156*** (0.040)	0.288*** (0.068)

Table A 3. Effects of heat stress on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.375*** (0.026)	0.295*** (0.029)	0.665*** (0.061)
Uganda	0.266*** (0.032)	0.139*** (0.036)	0.545*** (0.068)
Constant	3.035*** (0.140)	3.247*** (0.157)	2.872*** (0.260)
N second step	36319	30859	9024

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 4. Effects of floods on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Flood (1 yr)	0.012*** (0.002)	-0.005 (0.003)	0.010*** (0.003)
Young	-0.061** (0.031)	0.172*** (0.048)	-0.131*** (0.041)
Young # Flood (1 yr)	-0.015*** (0.003)	-0.007* (0.004)	0.002 (0.003)
Female	-0.426*** (0.013)	0.069*** (0.019)	-0.142*** (0.016)
Age of the respondent	0.055*** (0.005)	-0.029*** (0.007)	0.058*** (0.006)
Age of the respondent squared	-0.001*** (0.000)	0.000*** (0.000)	-0.001*** (0.000)
Respondent's years of education	0.013*** (0.002)	-0.030*** (0.003)	0.036*** (0.003)
Respondent's marital status	0.047** (0.019)	0.002 (0.029)	0.015 (0.025)
Respondent currently attends school	-0.581*** (0.022)	0.294*** (0.048)	-0.263*** (0.041)
Poor HH	0.042** (0.017)	0.174*** (0.034)	-0.182*** (0.029)
Female HH head	0.086*** (0.018)	-0.019 (0.033)	0.031 (0.029)
HH size	-0.018*** (0.003)	0.001 (0.004)	-0.009** (0.004)
Avg. age in HH	-0.058*** (0.003)	-0.020*** (0.006)	-0.006 (0.005)
Avg. age in HH sq.	0.001*** (0.000)	0.000*** (0.000)	0.000 (0.000)
Avg. education in HH	-0.014*** (0.004)	-0.051*** (0.006)	0.051*** (0.005)

Table A 4. Effects of floods on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	0.185*** (0.042)	0.014 (0.073)	-0.063 (0.064)
Females/males HH members	0.011* (0.006)	-0.021* (0.012)	0.034*** (0.010)
Share of working HH members	1.555*** (0.031)	0.836*** (0.045)	-0.460*** (0.037)
HH receives social protection	0.055*** (0.016)	0.115*** (0.030)	-0.040 (0.025)
Agricultural household	0.201*** (0.021)	0.816*** (0.037)	-0.568*** (0.036)
URCA	-0.005** (0.002)	0.005 (0.004)	-0.009*** (0.003)
Population density (log)	-0.038*** (0.008)	-0.052*** (0.015)	0.057*** (0.013)
Night-time lights (log)	-0.110*** (0.027)	-0.420*** (0.056)	0.301*** (0.052)
Long-term av. temp.	-0.022*** (0.003)	-0.016*** (0.006)	0.018*** (0.005)
Long-term temp. CV	-2.221*** (0.411)	-5.943*** (0.771)	3.421*** (0.651)
Long-term av. pre.	0.000 (0.004)	0.025*** (0.008)	-0.021*** (0.007)
Long-term pre. CV	-0.201*** (0.042)	-0.474*** (0.080)	0.506*** (0.071)
Malawi	0.268*** (0.023)	0.360*** (0.049)	-0.288*** (0.043)
Niger	1.086*** (0.055)	0.385*** (0.100)	-0.038 (0.083)
Nigeria	0.226*** (0.032)	-0.561*** (0.063)	0.549*** (0.055)

Table A 4. Effects of floods on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.141*** (0.026)	-0.268*** (0.051)	0.315*** (0.044)
Uganda	0.375*** (0.032)	-0.591*** (0.060)	0.636*** (0.052)
Constant	0.810*** (0.156)	2.681*** (0.284)	-2.709*** (0.243)
N first step	55709	36319	36319

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 5. Effects of floods on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Flood (1 yr)	-0.004** (0.002)	-0.006*** (0.002)	-0.011*** (0.003)
Young	-0.061*** (0.024)	-0.004 (0.026)	-0.215*** (0.050)
Young # Flood (1 yr)	-0.000 (0.002)	-0.000 (0.002)	0.016*** (0.004)
Female	-0.286*** (0.009)	-0.239*** (0.009)	-0.220*** (0.020)
Age of the respondent	0.028*** (0.003)	0.020*** (0.004)	-0.001 (0.007)
Age of the respondent squared	-0.000*** (0.000)	-0.000*** (0.000)	0.000 (0.000)
Respondent's years of education	0.011*** (0.002)	-0.004* (0.002)	0.017*** (0.003)
Respondent's marital status	-0.017 (0.015)	-0.022 (0.016)	-0.034 (0.028)
Respondent currently attends school	-0.542*** (0.022)	-0.485*** (0.023)	-0.509*** (0.055)
Poor HH	-0.012 (0.017)	0.008 (0.018)	-0.060* (0.032)
Female HH head	0.005 (0.017)	0.005 (0.019)	-0.039 (0.032)
HH size	-0.003 (0.002)	-0.000 (0.003)	-0.003 (0.004)
Avg. age in HH	-0.014*** (0.003)	-0.013*** (0.003)	0.003 (0.006)
Avg. age in HH sq.	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)
Avg. education in HH	0.003 (0.003)	-0.017*** (0.004)	0.014** (0.005)

Table A 5. Effects of floods on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	-0.083** (0.039)	-0.080* (0.043)	-0.095 (0.076)
Females/males HH members	0.011* (0.006)	-0.001 (0.007)	0.011 (0.009)
Share of working HH members	0.118*** (0.022)	0.176*** (0.024)	-0.077** (0.039)
HH receives social protection	0.022 (0.016)	0.026 (0.017)	-0.012 (0.029)
Agricultural household	-0.169*** (0.023)	-0.066** (0.031)	-0.193*** (0.029)
URCA	-0.006*** (0.002)	-0.003 (0.002)	-0.005 (0.004)
Population density (log)	-0.013* (0.008)	-0.035*** (0.009)	-0.001 (0.013)
Night-time lights (log)	0.189*** (0.029)	0.231*** (0.042)	0.077** (0.038)
Long-term av. temp.	-0.003 (0.003)	-0.009*** (0.003)	0.003 (0.006)
Long-term temp. CV	-0.041 (0.380)	-0.401 (0.420)	1.957*** (0.730)
Long-term av. pre.	-0.001 (0.004)	-0.001 (0.004)	0.013* (0.008)
Long-term pre. CV	0.150*** (0.045)	0.118** (0.050)	-0.010 (0.080)
Malawi	-0.517*** (0.026)	-0.461*** (0.029)	-0.233*** (0.060)
Niger	0.440*** (0.048)	0.409*** (0.054)	0.330*** (0.099)
Nigeria	-0.038 (0.035)	-0.150*** (0.040)	0.260*** (0.069)

Table A 5. Effects of floods on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.363*** (0.027)	0.306*** (0.029)	0.644*** (0.060)
Uganda	0.221*** (0.031)	0.131*** (0.035)	0.494*** (0.066)
Constant	3.102*** (0.144)	3.367*** (0.161)	3.022*** (0.265)
N second step	36319	30859	9024

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 6. Effects of droughts on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Drought (1 yr)	0.003 (0.021)	0.123*** (0.034)	-0.075*** (0.028)
Young	-0.153*** (0.028)	0.131*** (0.044)	-0.130*** (0.038)
Young # Drought (1 yr)	0.028 (0.032)	0.002 (0.057)	0.046 (0.047)
Female	-0.425*** (0.013)	0.069*** (0.019)	-0.141*** (0.016)
Age of the respondent	0.055*** (0.005)	-0.029*** (0.007)	0.058*** (0.006)
Age of the respondent squared	-0.001*** (0.000)	0.000*** (0.000)	-0.001*** (0.000)
Respondent's years of education	0.014*** (0.002)	-0.030*** (0.003)	0.036*** (0.003)
Respondent's marital status	0.046** (0.019)	0.006 (0.029)	0.010 (0.025)
Respondent currently attends school	-0.585*** (0.022)	0.290*** (0.048)	-0.261*** (0.040)
Poor HH	0.041** (0.017)	0.175*** (0.034)	-0.183*** (0.029)
Female HH head	0.087*** (0.018)	-0.015 (0.034)	0.029 (0.029)
HH size	-0.018*** (0.003)	0.001 (0.004)	-0.009** (0.004)
Avg. age in HH	-0.058*** (0.003)	-0.020*** (0.006)	-0.007 (0.005)
Avg. age in HH sq.	0.001*** (0.000)	0.000*** (0.000)	0.000 (0.000)
Avg. education in HH	-0.015*** (0.004)	-0.052*** (0.006)	0.051*** (0.005)

Table A 6. Effects of droughts on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	0.178*** (0.042)	0.011 (0.073)	-0.061 (0.064)
Females/males HH members	0.010* (0.006)	-0.020* (0.012)	0.033*** (0.010)
Share of working HH members	1.555*** (0.031)	0.831*** (0.045)	-0.451*** (0.037)
HH receives social protection	0.056*** (0.016)	0.115*** (0.030)	-0.039 (0.026)
Agricultural household	0.202*** (0.020)	0.813*** (0.037)	-0.564*** (0.036)
URCA	-0.005** (0.002)	0.004 (0.004)	-0.008** (0.003)
Population density (log)	-0.039*** (0.008)	-0.054*** (0.015)	0.058*** (0.013)
Night-time lights (log)	-0.115*** (0.027)	-0.411*** (0.056)	0.293*** (0.052)
Long-term av. temp.	-0.023*** (0.003)	-0.017*** (0.006)	0.017*** (0.005)
Long-term temp. CV	-2.410*** (0.406)	-5.734*** (0.757)	3.038*** (0.641)
Long-term av. pre.	0.001 (0.004)	0.023*** (0.008)	-0.019*** (0.007)
Long-term pre. CV	-0.249*** (0.040)	-0.424*** (0.076)	0.412*** (0.068)
Malawi	0.269*** (0.024)	0.315*** (0.050)	-0.253*** (0.043)
Niger	1.135*** (0.053)	0.332*** (0.095)	0.061 (0.079)
Nigeria	0.246*** (0.031)	-0.576*** (0.062)	0.583*** (0.054)

Table A 6. Effects of droughts on labour outcomes – youth vs adults – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.145*** (0.026)	-0.323*** (0.052)	0.363*** (0.045)
Uganda	0.378*** (0.032)	-0.608*** (0.060)	0.659*** (0.052)
Constant	0.970*** (0.152)	2.638*** (0.278)	-2.535*** (0.239)
N first step	55709	36319	36319

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 7. Effects of droughts on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Drought (1 yr)	0.038** (0.018)	0.060*** (0.019)	0.024 (0.034)
Young	-0.046** (0.022)	0.018 (0.024)	-0.063 (0.042)
Young # Drought (1 yr)	-0.082*** (0.026)	-0.097*** (0.026)	-0.251*** (0.068)
Female	-0.286*** (0.009)	-0.240*** (0.009)	-0.221*** (0.020)
Age of the respondent	0.029*** (0.003)	0.021*** (0.004)	-0.001 (0.007)
Age of the respondent squared	-0.000*** (0.000)	-0.000*** (0.000)	0.000 (0.000)
Respondent's years of education	0.010*** (0.002)	-0.004* (0.002)	0.017*** (0.003)
Respondent's marital status	-0.014 (0.015)	-0.019 (0.016)	-0.029 (0.028)
Respondent currently attends school	-0.541*** (0.022)	-0.486*** (0.023)	-0.506*** (0.054)
Poor HH	-0.011 (0.017)	0.008 (0.018)	-0.055* (0.032)
Female HH head	0.005 (0.017)	0.007 (0.019)	-0.037 (0.032)
HH size	-0.003 (0.003)	0.000 (0.003)	-0.002 (0.004)
Avg. age in HH	-0.013*** (0.003)	-0.013*** (0.003)	0.003 (0.006)
Avg. age in HH sq.	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)
Avg. education in HH	0.003 (0.003)	-0.017*** (0.004)	0.014** (0.005)

Table A 7. Effects of droughts on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	-0.083** (0.039)	-0.079* (0.043)	-0.098 (0.076)
Females/males HH members	0.011* (0.006)	-0.001 (0.007)	0.012 (0.009)
Share of working HH members	0.115*** (0.022)	0.170*** (0.024)	-0.088** (0.039)
HH receives social protection	0.021 (0.016)	0.025 (0.017)	-0.011 (0.029)
Agricultural household	-0.171*** (0.023)	-0.069** (0.031)	-0.195*** (0.029)
URCA	-0.006*** (0.002)	-0.004* (0.002)	-0.005 (0.004)
Population density (log)	-0.013* (0.008)	-0.035*** (0.009)	-0.001 (0.013)
Night-time lights (log)	0.192*** (0.029)	0.233*** (0.042)	0.084** (0.038)
Long-term av. temp.	-0.003 (0.003)	-0.009*** (0.003)	0.003 (0.006)
Long-term temp. CV	0.095 (0.377)	-0.138 (0.416)	2.159*** (0.725)
Long-term av. pre.	-0.001 (0.004)	-0.001 (0.004)	0.010 (0.008)
Long-term pre. CV	0.182*** (0.042)	0.175*** (0.047)	0.055 (0.076)
Malawi	-0.526*** (0.027)	-0.480*** (0.030)	-0.225*** (0.060)
Niger	0.407*** (0.046)	0.347*** (0.051)	0.275*** (0.094)
Nigeria	-0.049 (0.034)	-0.170*** (0.040)	0.241*** (0.067)

Table A 7. Effects of droughts on labour outcomes – youth vs adults – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.348*** (0.027)	0.279*** (0.029)	0.643*** (0.060)
Uganda	0.212*** (0.031)	0.116*** (0.035)	0.483*** (0.066)
Constant	3.035*** (0.140)	3.242*** (0.157)	2.867*** (0.260)
N second step	36319	30859	9024

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 8. Effects of heat stress on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Heat stress (1 yr)	0.009*** (0.003)	-0.020*** (0.006)	0.021*** (0.005)
Female	-0.256*** (0.029)	0.063 (0.058)	-0.096** (0.049)
Female # Heat stress (1 yr)	-0.027*** (0.004)	-0.021*** (0.007)	0.017*** (0.006)
Age of the respondent	-0.148*** (0.052)	0.044 (0.098)	0.003 (0.085)
Age of the respondent squared	0.005*** (0.001)	-0.002 (0.002)	0.001 (0.002)
Respondent's years of education	-0.003 (0.004)	-0.011 (0.008)	0.019*** (0.007)
Respondent's marital status	-0.001 (0.032)	0.256*** (0.057)	-0.155*** (0.049)
Respondent currently attends school	-0.609*** (0.025)	0.306*** (0.056)	-0.270*** (0.047)
Poor HH	0.028 (0.029)	0.290*** (0.063)	-0.259*** (0.054)
Female HH head	0.063*** (0.024)	0.025 (0.052)	0.007 (0.044)
HH size	-0.005 (0.004)	0.022*** (0.008)	-0.021*** (0.007)
Avg. age in HH	-0.068*** (0.007)	-0.004 (0.014)	-0.012 (0.012)
Avg. age in HH sq.	0.001*** (0.000)	0.000 (0.000)	0.000 (0.000)
Avg. education in HH	-0.022*** (0.006)	-0.038*** (0.012)	0.032*** (0.010)

Table A 8. Effects of heat stress on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	0.431*** (0.071)	0.011 (0.150)	-0.083 (0.128)
Females/males HH members	0.009 (0.010)	-0.054** (0.024)	0.040** (0.020)
Share of working HH members	1.875*** (0.045)	0.865*** (0.082)	-0.534*** (0.069)
HH receives social protection	0.063** (0.025)	0.117** (0.056)	-0.078* (0.047)
Agricultural household	0.220*** (0.036)	0.672*** (0.067)	-0.434*** (0.063)
URCA	-0.006* (0.003)	0.004 (0.007)	-0.005 (0.006)
Population density (log)	-0.062*** (0.013)	-0.023 (0.029)	0.019 (0.026)
Night-time lights (log)	-0.099* (0.055)	-0.462*** (0.109)	0.401*** (0.107)
Long-term avg. temp.	-0.024*** (0.005)	-0.006 (0.011)	-0.007 (0.010)
Long-term temp. CV	-2.080*** (0.679)	-2.146 (1.398)	-0.446 (1.210)
Long-term avg. pre.	-0.001 (0.007)	0.050*** (0.015)	-0.051*** (0.012)
Long-term pre. CV	-0.094 (0.068)	-0.106 (0.144)	0.198 (0.127)
Malawi	0.077* (0.041)	0.213** (0.093)	-0.163** (0.081)
Niger	0.576*** (0.094)	-0.304* (0.180)	0.414*** (0.156)
Nigeria	-0.160*** (0.054)	-0.746*** (0.116)	0.681*** (0.102)

Table A 8. Effects of heat stress on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	-0.147*** (0.043)	-0.269*** (0.092)	0.255*** (0.080)
Uganda	0.145*** (0.052)	-0.637*** (0.109)	0.584*** (0.096)
Constant	2.838*** (0.541)	1.071 (1.067)	-0.968 (0.915)
N first step	21 392	11 186	11 186

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 9. Effects of heat stress on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Heat stress (1 yr)	0.013*** (0.003)	0.010*** (0.003)	0.022*** (0.007)
Female	-0.223*** (0.024)	-0.215*** (0.025)	-0.085 (0.069)
Female # Heat stress (1 yr)	-0.009*** (0.003)	-0.012*** (0.004)	-0.010 (0.009)
Age of the respondent	0.037 (0.043)	0.046 (0.045)	0.232* (0.119)
Age of the respondent squared	-0.000 (0.001)	-0.001 (0.001)	-0.005* (0.003)
Respondent's years of education	0.008** (0.004)	0.004 (0.004)	0.012 (0.009)
Respondent's marital status	-0.034 (0.027)	0.018 (0.029)	-0.150** (0.059)
Respondent currently attends school	-0.529*** (0.025)	-0.473*** (0.026)	-0.540*** (0.072)
Poor HH	0.016 (0.029)	0.031 (0.030)	0.009 (0.069)
Female HH head	0.017 (0.025)	0.026 (0.026)	-0.090 (0.058)
HH size	-0.000 (0.004)	0.006 (0.004)	-0.015 (0.010)
Avg. age in HH	-0.022*** (0.007)	-0.024*** (0.007)	-0.002 (0.016)
Avg. age in HH sq.	0.000*** (0.000)	0.000*** (0.000)	0.000 (0.000)
Avg. education in HH	-0.007 (0.006)	-0.019*** (0.006)	0.008 (0.013)

Table A 9. Effects of heat stress on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	-0.113 (0.072)	-0.122* (0.074)	-0.031 (0.191)
Females/males HH members	0.012 (0.011)	0.004 (0.011)	0.024 (0.022)
Share of working HH members	0.087** (0.040)	0.132*** (0.042)	0.027 (0.082)
HH receives social protection	0.017 (0.026)	0.027 (0.028)	0.019 (0.070)
Agricultural household	-0.152*** (0.042)	-0.057 (0.050)	-0.252*** (0.069)
URCA	-0.003 (0.003)	-0.003 (0.004)	0.003 (0.008)
Population density (log)	-0.026* (0.014)	-0.042*** (0.015)	0.024 (0.032)
Night-time lights (log)	0.242*** (0.066)	0.233*** (0.076)	0.049 (0.116)
Long-term avg. temp.	-0.014*** (0.005)	-0.014** (0.006)	-0.007 (0.013)
Long-term temp. CV	-0.229 (0.660)	0.356 (0.697)	0.626 (1.577)
Long-term avg. pre.	-0.015** (0.006)	-0.010 (0.007)	0.018 (0.016)
Long-term pre. CV	0.211*** (0.071)	0.253*** (0.076)	0.058 (0.168)
Malawi	-0.663*** (0.045)	-0.659*** (0.048)	-0.412*** (0.121)
Niger	0.346*** (0.083)	0.205** (0.087)	0.662*** (0.206)
Nigeria	-0.108* (0.060)	-0.230*** (0.066)	0.375*** (0.140)

Table A 9. Effects of heat stress on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.239*** (0.044)	0.181*** (0.047)	0.648*** (0.120)
Uganda	0.143** (0.056)	0.056 (0.059)	0.677*** (0.138)
Constant	3.219*** (0.475)	3.076*** (0.498)	0.291 (1.300)
N second step	11 186	9 994	1 807

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own calculations

Table A 10. Effects of floods on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Flood (1 yr)	0.005 (0.003)	-0.014*** (0.005)	0.020*** (0.005)
Female	-0.498*** (0.031)	-0.194*** (0.058)	0.106** (0.050)
Female # Flood (1 yr)	0.016*** (0.004)	0.022*** (0.006)	-0.017*** (0.005)
Age of the respondent	-0.143*** (0.052)	0.066 (0.097)	-0.015 (0.085)
Age of the respondent squared	0.005*** (0.001)	-0.003 (0.002)	0.002 (0.002)
Respondent's years of education	-0.002 (0.004)	-0.012 (0.008)	0.020*** (0.007)
Respondent's marital status	0.003 (0.032)	0.256*** (0.057)	-0.150*** (0.049)
Respondent currently attends school	-0.608*** (0.025)	0.307*** (0.056)	-0.273*** (0.047)
Poor HH	0.028 (0.029)	0.296*** (0.062)	-0.264*** (0.054)
Female HH head	0.061** (0.024)	0.020 (0.052)	0.013 (0.044)
HH size	-0.005 (0.004)	0.022*** (0.008)	-0.022*** (0.007)
Avg. age in HH	-0.067*** (0.007)	-0.006 (0.014)	-0.009 (0.012)
Avg. age in HH sq.	0.001*** (0.000)	0.000 (0.000)	0.000 (0.000)
Avg. education in HH	-0.024*** (0.006)	-0.035*** (0.012)	0.028*** (0.010)
Share of dependent HH members (<15 and >64)	0.428*** (0.071)	0.013 (0.149)	-0.087 (0.128)

Table A 10. Effects of floods on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Females/males HH members	0.012 (0.010)	-0.051** (0.024)	0.038* (0.020)
Share of working HH members	1.865*** (0.045)	0.887*** (0.081)	-0.564*** (0.069)
HH receives social protection	0.062** (0.025)	0.122** (0.056)	-0.086* (0.047)
Agricultural household	0.214*** (0.036)	0.669*** (0.066)	-0.437*** (0.062)
URCA	-0.006* (0.003)	0.006 (0.007)	-0.008 (0.006)
Population density (log)	-0.063*** (0.013)	-0.032 (0.030)	0.029 (0.027)
Night-time lights (log)	-0.083 (0.055)	-0.467*** (0.109)	0.414*** (0.107)
Long-term av. temp.	-0.023*** (0.005)	-0.011 (0.011)	-0.001 (0.009)
Long-term temp. CV	-1.787*** (0.680)	-3.651*** (1.393)	1.411 (1.207)
Long-term av. pre.	-0.003 (0.007)	0.044*** (0.015)	-0.045*** (0.012)
Long-term pre. CV	0.019 (0.071)	-0.210 (0.148)	0.373*** (0.133)
Malawi	0.083** (0.041)	0.351*** (0.092)	-0.320*** (0.079)
Niger	0.484*** (0.096)	-0.000 (0.179)	0.026 (0.156)
Nigeria	-0.187*** (0.054)	-0.588*** (0.114)	0.493*** (0.101)
United Republic of Tanzania	-0.169*** (0.043)	-0.184** (0.091)	0.146* (0.080)

Table A 10. Effects of floods on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Uganda	0.145*** (0.051)	-0.458*** (0.105)	0.394*** (0.093)
Constant	2.648*** (0.542)	1.082 (1.066)	-1.159 (0.913)
N first step	21 392	11 186	11 186

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own calculations

Table A 11. Effects of floods on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Flood (1 yr)	-0.002 (0.003)	-0.007** (0.003)	-0.002 (0.006)
Female	-0.313*** (0.025)	-0.341*** (0.026)	-0.137** (0.070)
Female # Flood (1 yr)	0.007*** (0.003)	0.011*** (0.003)	-0.001 (0.007)
Age of the respondent	0.041 (0.043)	0.053 (0.045)	0.222* (0.120)
Age of the respondent squared	-0.000 (0.001)	-0.001 (0.001)	-0.005 (0.003)
Respondent's years of education	0.009** (0.004)	0.004 (0.004)	0.012 (0.009)
Respondent's marital status	-0.031 (0.027)	0.020 (0.029)	-0.153*** (0.059)
Respondent currently attends school	-0.528*** (0.025)	-0.471*** (0.026)	-0.540*** (0.072)
Poor HH	0.016 (0.029)	0.032 (0.030)	0.005 (0.069)
Female HH head	0.021 (0.025)	0.029 (0.026)	-0.086 (0.058)
HH size	0.000 (0.004)	0.006 (0.004)	-0.016* (0.010)
Avg. age in HH	-0.022*** (0.007)	-0.025*** (0.007)	0.000 (0.016)
Avg. age in HH sq.	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)
Avg. education in HH	-0.008 (0.006)	-0.019*** (0.006)	0.007 (0.013)

Table A 11. Effects of floods on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	-0.119* (0.072)	-0.127* (0.074)	-0.012 (0.192)
Females/males HH members	0.011 (0.011)	0.004 (0.011)	0.024 (0.022)
Share of working HH members	0.075* (0.040)	0.126*** (0.042)	0.015 (0.082)
HH receives social protection	0.017 (0.026)	0.028 (0.028)	0.027 (0.071)
Agricultural household	-0.149*** (0.042)	-0.050 (0.050)	-0.257*** (0.069)
URCA	-0.004 (0.003)	-0.004 (0.004)	0.004 (0.008)
Population density (log)	-0.024* (0.014)	-0.041*** (0.015)	0.038 (0.032)
Night-time lights (log)	0.243*** (0.067)	0.237*** (0.075)	0.018 (0.116)
Long-term av. temp.	-0.013** (0.005)	-0.013** (0.006)	-0.004 (0.013)
Long-term temp. CV	0.268 (0.657)	0.487 (0.697)	1.463 (1.582)
Long-term av. pre.	-0.013** (0.006)	-0.008 (0.007)	0.017 (0.017)
Long-term pre. CV	0.244*** (0.076)	0.248*** (0.081)	0.042 (0.176)
Malawi	-0.715*** (0.044)	-0.683*** (0.046)	-0.464*** (0.122)
Niger	0.254*** (0.084)	0.183** (0.089)	0.541** (0.211)
Nigeria	-0.155** (0.060)	-0.248*** (0.066)	0.317** (0.144)

Table A 11. Effects of floods on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.213*** (0.045)	0.172*** (0.047)	0.616*** (0.121)
Uganda	0.090* (0.054)	0.033 (0.057)	0.581*** (0.134)
Constant	3.208*** (0.477)	3.116*** (0.500)	0.394 (1.300)
N second step	11 186	9 994	1 807

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 12. Effects of droughts on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Drought (1 yr)	-0.049 (0.035)	-0.010 (0.073)	0.043 (0.060)
Female	-0.416*** (0.024)	-0.081* (0.044)	0.017 (0.038)
Female # Drought (1 yr)	0.095** (0.048)	0.212** (0.099)	-0.136* (0.080)
Age of the respondent	-0.141*** (0.052)	0.062 (0.097)	-0.010 (0.084)
Age of the respondent squared	0.005*** (0.001)	-0.003 (0.002)	0.002 (0.002)
Respondent's years of education	-0.002 (0.004)	-0.010 (0.008)	0.019*** (0.007)
Respondent's marital status	-0.003 (0.032)	0.261*** (0.056)	-0.155*** (0.048)
Respondent currently attends school	-0.608*** (0.025)	0.307*** (0.056)	-0.270*** (0.046)
Poor HH	0.024 (0.029)	0.291*** (0.062)	-0.262*** (0.054)
Female HH head	0.061** (0.024)	0.014 (0.052)	0.018 (0.044)
HH size	-0.005 (0.004)	0.022*** (0.008)	-0.021*** (0.007)
Avg. age in HH	-0.067*** (0.007)	-0.004 (0.014)	-0.012 (0.012)
Avg. age in HH sq.	0.001*** (0.000)	0.000 (0.000)	0.000 (0.000)
Avg. education in HH	-0.023*** (0.006)	-0.037*** (0.012)	0.030*** (0.010)

Table A 12. Effects of droughts on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	0.424*** (0.071)	-0.002 (0.148)	-0.078 (0.128)
Females/males HH members	0.012 (0.010)	-0.049** (0.024)	0.036* (0.020)
Share of working HH members	1.867*** (0.045)	0.874*** (0.081)	-0.550*** (0.069)
HH receives social protection	0.068*** (0.025)	0.121** (0.056)	-0.082* (0.047)
Agricultural household	0.220*** (0.036)	0.659*** (0.066)	-0.424*** (0.062)
URCA	-0.006* (0.003)	0.005 (0.007)	-0.006 (0.006)
Population density (log)	-0.065*** (0.013)	-0.034 (0.030)	0.031 (0.027)
Night-time lights (log)	-0.095* (0.055)	-0.455*** (0.108)	0.398*** (0.107)
Long-term av. temp.	-0.025*** (0.005)	-0.011 (0.011)	-0.002 (0.009)
Long-term temp. CV	-2.334*** (0.670)	-3.479** (1.362)	0.882 (1.180)
Long-term av. pre.	-0.003 (0.007)	0.042*** (0.015)	-0.045*** (0.012)
Long-term pre. CV	-0.102 (0.067)	-0.183 (0.142)	0.258** (0.126)
Malawi	0.102** (0.041)	0.312*** (0.092)	-0.290*** (0.080)
Niger	0.622*** (0.092)	-0.033 (0.171)	0.149 (0.149)
Nigeria	-0.135** (0.053)	-0.600*** (0.112)	0.538*** (0.099)

Table A 12. Effects of droughts on labour outcomes – young women vs young men – first step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	-0.132*** (0.043)	-0.219** (0.092)	0.189** (0.080)
Uganda	0.172*** (0.050)	-0.470*** (0.105)	0.417*** (0.092)
Constant	2.859*** (0.540)	1.028 (1.059)	-0.935 (0.907)
N first step	21 392	11 186	11 186

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Source: authors' own elaboration

Table A 13. Effects of droughts on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Drought (1 yr)	0.002 (0.032)	0.008 (0.033)	-0.067 (0.085)
Female	-0.265*** (0.020)	-0.272*** (0.021)	-0.134** (0.052)
Female # Drought (1 yr)	-0.019 (0.040)	-0.015 (0.042)	-0.056 (0.116)
Age of the respondent	0.041 (0.043)	0.053 (0.045)	0.223* (0.120)
Age of the respondent squared	-0.000 (0.001)	-0.001 (0.001)	-0.005 (0.003)
Respondent's years of education	0.009** (0.004)	0.004 (0.004)	0.012 (0.009)
Respondent's marital status	-0.032 (0.027)	0.019 (0.029)	-0.153** (0.059)
Respondent currently attends school	-0.528*** (0.025)	-0.471*** (0.026)	-0.540*** (0.072)
Poor HH	0.014 (0.029)	0.030 (0.030)	0.010 (0.069)
Female HH head	0.019 (0.025)	0.027 (0.026)	-0.087 (0.058)
HH size	-0.000 (0.004)	0.006 (0.004)	-0.016 (0.010)
Avg. age in HH	-0.022*** (0.007)	-0.024*** (0.007)	0.001 (0.016)
Avg. age in HH sq.	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)
Avg. education in HH	-0.008 (0.006)	-0.020*** (0.006)	0.007 (0.013)

Table A 13. Effects of droughts on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
Share of dependent HH members (<15 and >64)	-0.119* (0.072)	-0.129* (0.074)	-0.011 (0.192)
Females/males HH members	0.012 (0.011)	0.004 (0.011)	0.024 (0.022)
Share of working HH members	0.074* (0.040)	0.122*** (0.042)	0.011 (0.082)
HH receives social protection	0.017 (0.027)	0.028 (0.028)	0.030 (0.071)
Agricultural household	-0.150*** (0.042)	-0.054 (0.050)	-0.255*** (0.069)
URCA	-0.004 (0.003)	-0.004 (0.004)	0.003 (0.008)
Population density (log)	-0.024* (0.014)	-0.041*** (0.015)	0.037 (0.032)
Night-time lights (log)	0.244*** (0.067)	0.239*** (0.075)	0.030 (0.115)
Long-term av. temp.	-0.014** (0.005)	-0.014** (0.006)	-0.004 (0.013)
Long-term temp. CV	0.215 (0.651)	0.574 (0.690)	1.630 (1.552)
Long-term av. pre.	-0.013** (0.006)	-0.009 (0.007)	0.018 (0.017)
Long-term pre. CV	0.234*** (0.071)	0.267*** (0.076)	0.070 (0.169)
Malawi	-0.711*** (0.044)	-0.688*** (0.047)	-0.438*** (0.122)
Niger	0.266*** (0.080)	0.165* (0.085)	0.503** (0.202)
Nigeria	-0.149** (0.059)	-0.249*** (0.065)	0.301** (0.141)

Table A 13. Effects of droughts on labour outcomes – young women vs young men – second step

	(1) Labour	(2) Agricultural labour	(3) Non-agricultural labour
United Republic of Tanzania	0.219*** (0.045)	0.168*** (0.047)	0.634*** (0.121)
Uganda	0.092* (0.054)	0.029 (0.057)	0.584*** (0.134)
Constant	3.206*** (0.475)	3.050*** (0.498)	0.297 (1.291)
N second step	11 186	9 994	1 807

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. In the second step, the outcome variable (working time) is log-transformed for the regression estimation. HH = household, avg. = average, URCA = Urban-Rural Catchment Area, CV = coefficient of variation

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Disaggregation by sub-sectors

Table A 14. Effects of heat stress on young female vs young male workers' likelihood to work and weekly working hours in sub-sectors of agriculture and non-agriculture (conditional average marginal effects)

	Agriculture		Non-agriculture	
	HH farm labour	Other agricultural labour	Non-farm HH business	Other non-agricultural labour
<i>First step (probit)</i>				
Female	0.0553*** (0.0105)	-0.0776*** (0.0092)	0.0204** (0.0088)	-0.0375*** (0.0067)
Heat stress	-0.0032*** (0.0010)	0.0006 (0.0008)	0.0049*** (0.0008)	0.0002 (0.0007)
Female x Heat stress	-0.0052*** (0.0014)	0.0010 (0.0012)	0.0008 (0.0011)	0.0018** (0.0009)
Average net effect of heat stress on young women	-0.0084*** (0.0011)	0.0016 (0.0010)	0.0057*** (0.0008)	0.0020*** (0.0007)
<i>Second step (OLS)</i>				
Female	-166.0847*** (31.3209)	-307.0338*** (63.7399)	-50.3647 (144.1408)	-84.2899 (248.4398)
Heat stress	11.7462*** (3.2124)	4.9466 (5.1496)	27.4611** (12.9996)	73.1582*** (25.9537)
Female x Heat stress	-16.2560*** (4.4970)	0.9793 (7.8233)	-24.5779 (16.9585)	-4.2257 (36.9976)
Average net effect of heat stress on young women	-4.5098 (3.6076)	5.9259 (6.6981)	2.8832 (6.6109)	68.9325** (12.8075)
N first step	11 186	11 186	11 186	11 186
N second step	9 041	1 631	1 168	658
Mean outcome second step (hours)	18.94	15.53	25.94	36.36

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 15. Effects of extreme precipitation on young female vs young male workers' likelihood to work and weekly working hours in sub-sectors of agriculture and non-agriculture (conditional average marginal effects)

	Agriculture		Non-agriculture	
	HH farm labour	Other agricultural labour	Non-farm HH business	Other non-agricultural labour
<i>First step (probit)</i>				
Female	0.0146 (0.0108)	-0.0712*** (0.0091)	0.0330*** (0.0086)	-0.0187*** (0.0070)
Extreme precipitation	-0.0011 (0.0011)	0.0011 (0.0010)	0.0030*** (0.0009)	0.0008 (0.0005)
Female x Extreme precipitation	0.0025** (0.0012)	-0.0003 (0.0012)	-0.0015 (0.0009)	-0.0014** (0.0007)
Average net effect extreme precipitation on young women	0.0014 (0.0011)	0.0007 (0.0011)	0.0016* (0.0008)	-0.0006 (0.0006)
<i>Second step (OLS)</i>				
Female	-313.2918*** (32.7248)	-323.9997*** (59.6105)	-204.4517 (133.6461)	-200.4822 (276.0542)
Extreme precipitation	-7.0393** (2.6557)	-22.0040** (8.9635)	-10.1328 (12.2017)	29.1677 (18.1714)
Female x Extreme precipitation	10.3694*** (3.1013)	5.9058 (8.1843)	-0.8937 (13.9320)	12.6589 (22.2098)
Average net effect of extreme precipitation on young women	3.3301 (2.8396)	-16.0982** (6.8227)	-11.0265 (12.5154)	41.8266* (22.4666)
N first step	11 186	11 186	11 186	11 186
N second step	9 041	1 631	1 168	658
Mean outcome second step (hours)	18.94	15.53	25.94	36.36

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

Table A 16. Effects of droughts on young female vs young male workers' likelihood to work and weekly working hours in sub-sectors of agriculture and non-agriculture (conditional average marginal effects)

	Agriculture		Non-agriculture	
	HH farm labour	Other agricultural labour	Non-farm HH business	Other non-agricultural labour
<i>First step (probit)</i>				
Female	0.0197** (0.0087)	-0.0783*** (0.0079)	0.0276*** (0.0070)	-0.0292*** (0.0054)
Drought	0.0076 (0.0114)	-0.0233*** (0.0089)	0.0011 (0.0111)	0.0051 (0.0070)
Female x Drought	0.0425** (0.0168)	0.0182 (0.0130)	-0.0221 (0.0147)	0.0016 (0.0105)
Average net effect droughts on young women	0.0501*** (0.0135)	-0.0051 (0.0100)	-0.0210* (0.0111)	0.0066 (0.0091)
<i>Second step (OLS)</i>				
Female	-250.1297*** (26.3338)	-273.9890*** (57.9930)	-192.5175* (100.8082)	-72.6893 (194.8574)
Drought	-3.9620 (38.7541)	61.1633 (56.5682)	-75.0235 (171.4399)	-192.9457 (272.0631)
Female x Drought	26.0273 (50.8228)	-87.6970 (88.1385)	-1.1e+02 (232.5904)	-71.4047 (423.7989)
Average net effect of droughts on young women	22.0653 (38.9399)	-26.5337 (70.2113)	-188.6362 (187.6072)	-264.3504 (363.3657)
N first step	11 186	11 186	11 186	11 186
N second step	9 041	1 631	1 168	658
Mean outcome second step (hours)	18.94	15.53	25.94	36.36

Notes: The first step is estimated using a probit model and the second step is estimated using a linear regression model. The effects shown are conditional average marginal effects and can be interpreted in percentage points (first step) and in minutes (second step). In the second step, the outcome variable (working time) is log-transformed for the regression estimation. During the calculation of average marginal effects, Duan smearing retransformation is applied.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: authors' own elaboration

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